Huawei Cloud · Documentation Guide

Cloud Secret Management Service / Cross-Cloud Replication

HUAWEI CLOUD · LAB GUIDE

Replicating Rotated AWS Secrets Manager Secrets to Huawei Cloud DEW CSMS

Document Type	Lab Guide
Lab Topic	Cross-Cloud Secret Replication
AWS Region	us-east-1 (US East — N. Virginia)
Huawei Region	af-south-1 (AF-Johannesburg)
AWS Service	Secrets Manager · Lambda · EventBridge · CloudTrail
Huawei Service	DEW · Cloud Secret Management Service (CSMS)
Author	Oluwatoba Daniel Mapayi
Lab Level	Intermediate
Reviewed by	Abdurahman Olusokun

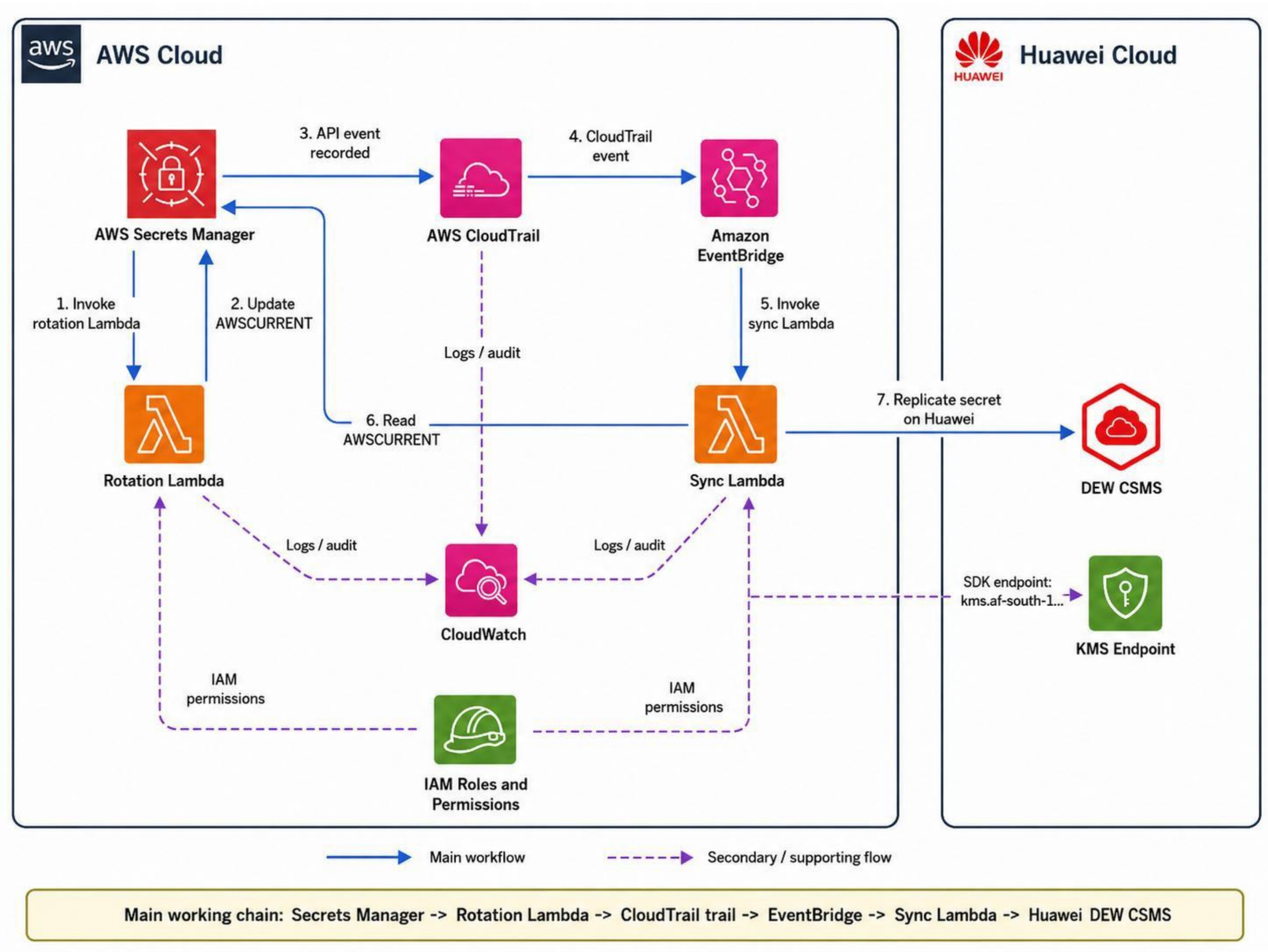
Overview

This lab walks you through a complete, end-to-end system that automatically copies a rotated secret from AWS Secrets Manager into Huawei Cloud's Data Encryption Workshop (DEW) — specifically its Cloud Secret Management Service (CSMS).

By the end of this lab, you will have built a working pipeline that rotates a secret in AWS every four hours and automatically replicates the latest value to a Huawei Cloud secret — across two cloud providers, two regions, and two security systems.

This lab is for cloud engineers and architects who are comfortable with the AWS console and have basic familiarity with Python and AWS Lambda.

What you will build



Before You Begin

Gather everything in the list below before you start the first step.

AWS requirements

- An AWS account with console access and permission to create Secrets Manager secrets, Lambda functions, IAM roles and inline policies, EventBridge rules, and CloudTrail trails.
- Access to AWS CloudShell (available in the AWS console — no local tools required).
- AWS region set to: us-east-1 (US East — N. Virginia).

Huawei Cloud requirements

- A Huawei Cloud account with access to the AF-Johannesburg region (af-south-1).
- An Access Key (AK) and Secret Key (SK) for a Huawei IAM user with permission to create and manage CSMS secrets.
- The Huawei Project ID for the AF-Johannesburg project — find this in the Huawei console under My Credentials.

Key resource names used in this lab

AWS source secret	lab/aws-to-huawei/source-secret
AWS rotation Lambda	rotate-aws-to-huawei-source-secret
AWS sync Lambda	sync-aws-secret-to-huawei-csms
Huawei target secret	aws-huawei_replica
EventBridge rule	sync-huawei-after-aws-secret-rotation
CloudTrail trail	secret-rotation-eventbridge-trail
Huawei CSMS endpoint	https://kms.af-south-1.myhuaweicloud.com

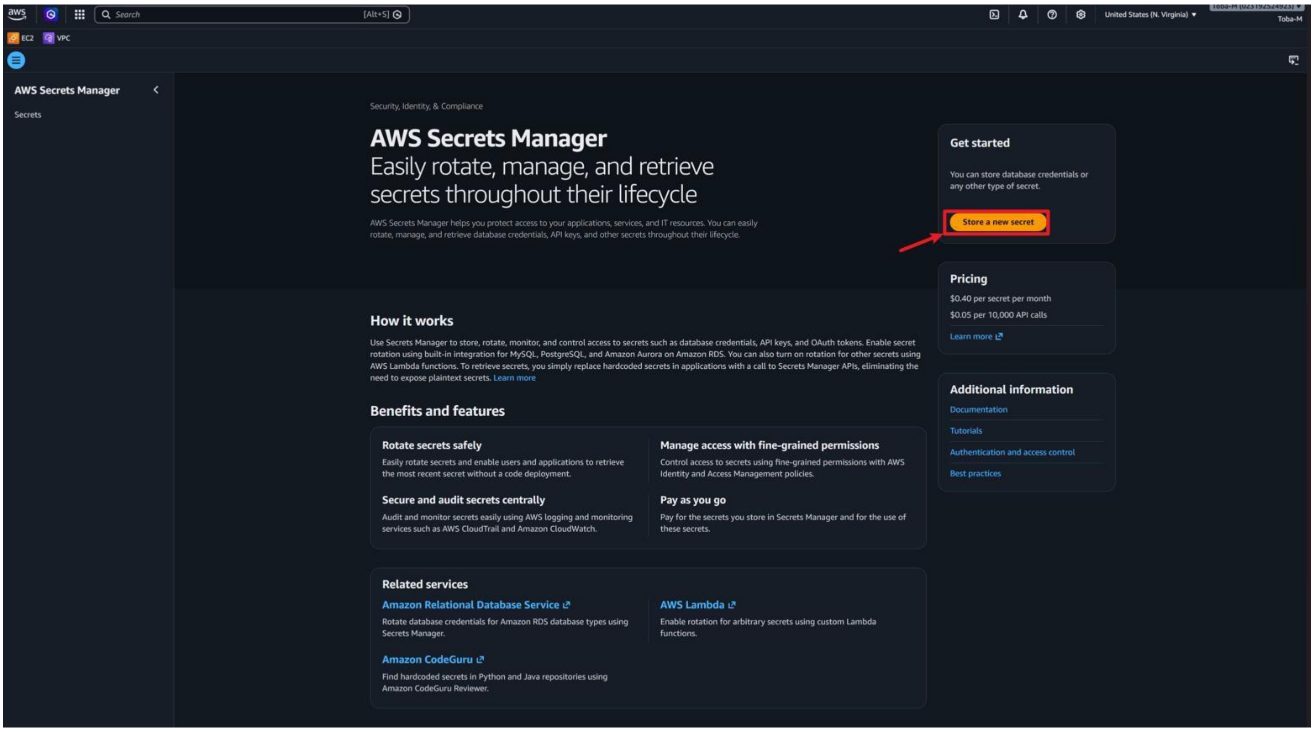
WARNING — Wrong endpoint
The Huawei CSMS endpoint is NOT https://csms.af-south-1.myhuaweicloud.com.
Use https://kms.af-south-1.myhuaweicloud.com — this is the correct endpoint for the Huawei CSMS SDK in AF-Johannesburg.
Using the wrong endpoint produces a HostUnreachableException error.

STAGE 1
CREATE THE AWS SOURCE SECRET

Step 1 — Open AWS Secrets Manager

Open the **AWS Console** and search for **Secrets Manager**.

Click **Secrets Manager** in the search results to open the service.



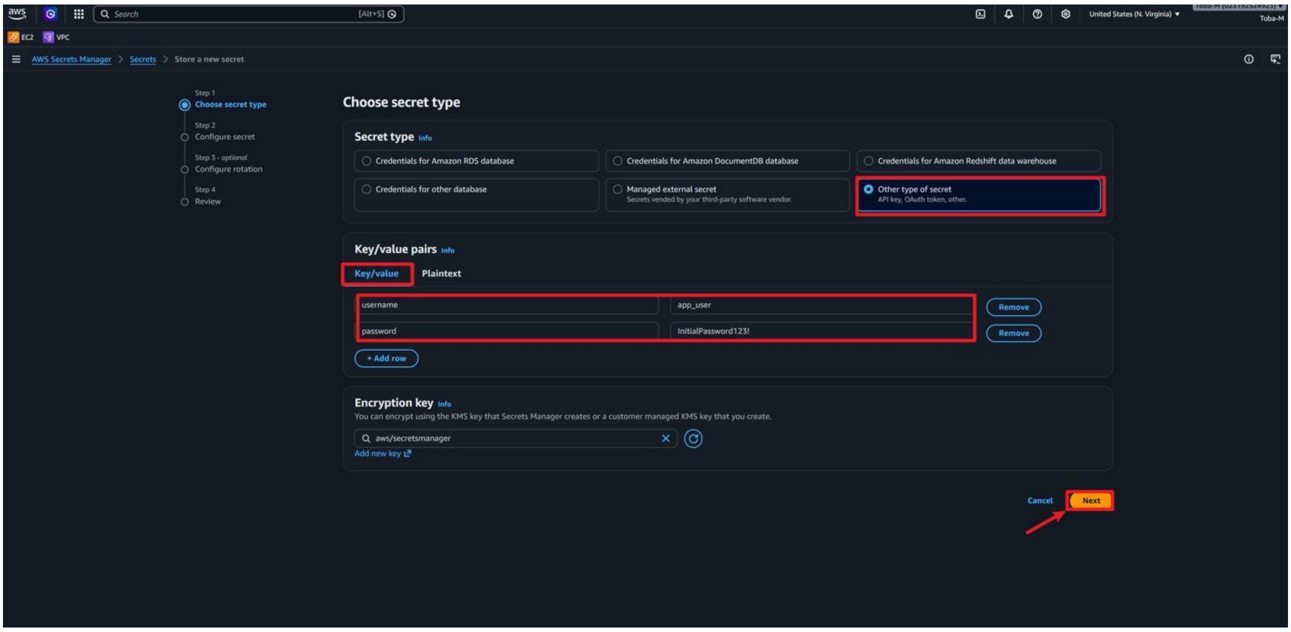
Step 2 — Start creating a new secret

Click the button labelled **Store a new secret**.

Step 3 — Choose the secret type

On the secret type page, choose **Other type of secret**.

Do not choose "Credentials for Amazon RDS database" — that option is designed for database passwords, not the custom key-value secret this lab uses.



Step 4 — Enter the secret key-value pairs

Under **Key/value pairs**, add the first key.

Set the key to `username` and set the value to `app_user`.

Click **Add row** to add a second pair.

Set the second key to `password` and set the value to `InitialPassword123!`.

Leave the encryption key as the default: **aws/secretsmanager**.

Click **Next**.

Step 5 — Name the secret

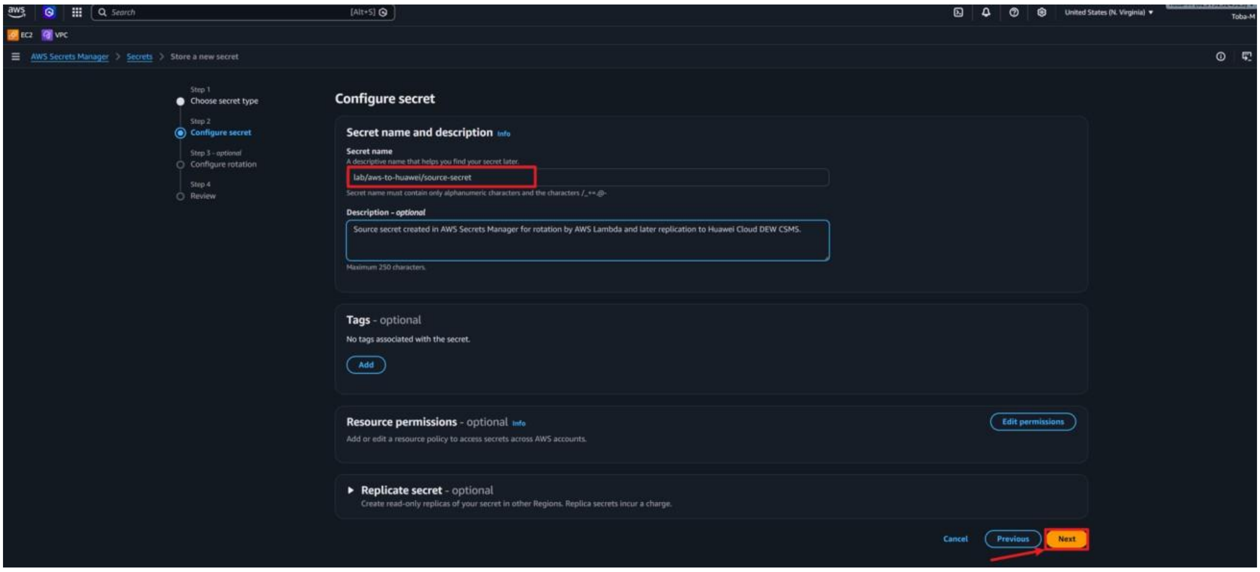
Set the secret name to:

```
lab/aws-to-huawei/source-secret
```

Set the description to:

```
Source secret created in AWS Secrets Manager for rotation by AWS Lambda and later
replication to Huawei Cloud DEW CSMS.
```

Click **Next**.

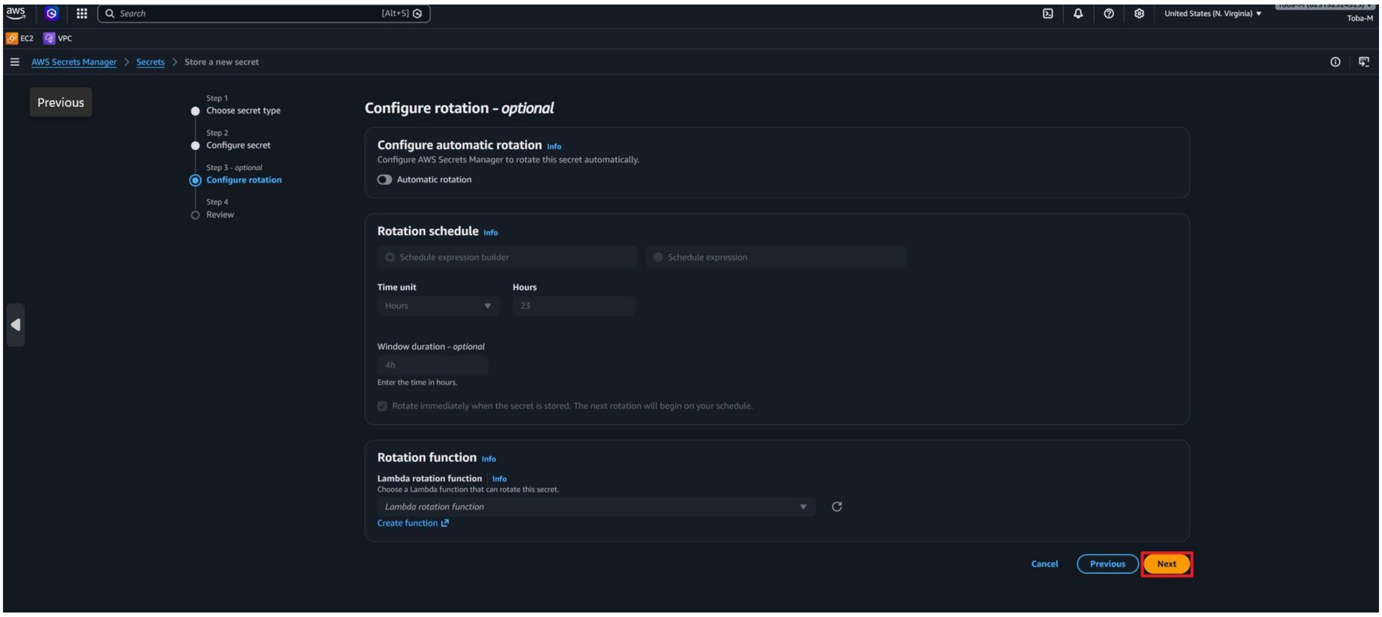


Step 6 — Skip rotation for now

On the automatic rotation page, leave rotation disabled.

Click **Next**.

> **NOTE**
> You will enable rotation in Stage 3. Skipping it here keeps the setup simple.



Step 7 — Review and store

Review the configuration on the final page.

Click **Store**.

If this worked, you should see: the secret `lab/aws-to-huawei/source-secret` in the Secrets Manager list with a green success banner.

STAGE 2
CREATE THE ROTATION LAMBDA

Step 1 — Create the Lambda function

Go to **AWS Lambda** → **Functions** → **Create function**.

Choose **Author from scratch**.

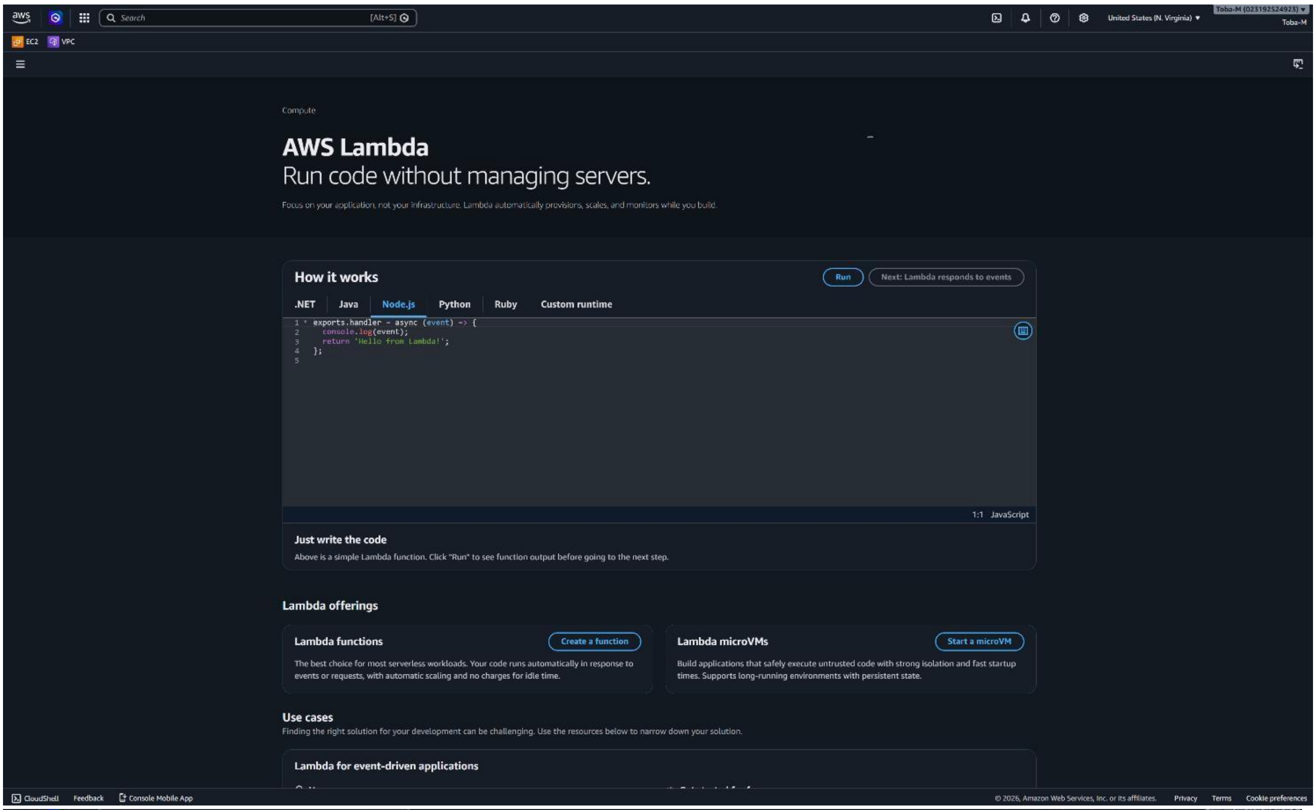
Set the function name to:

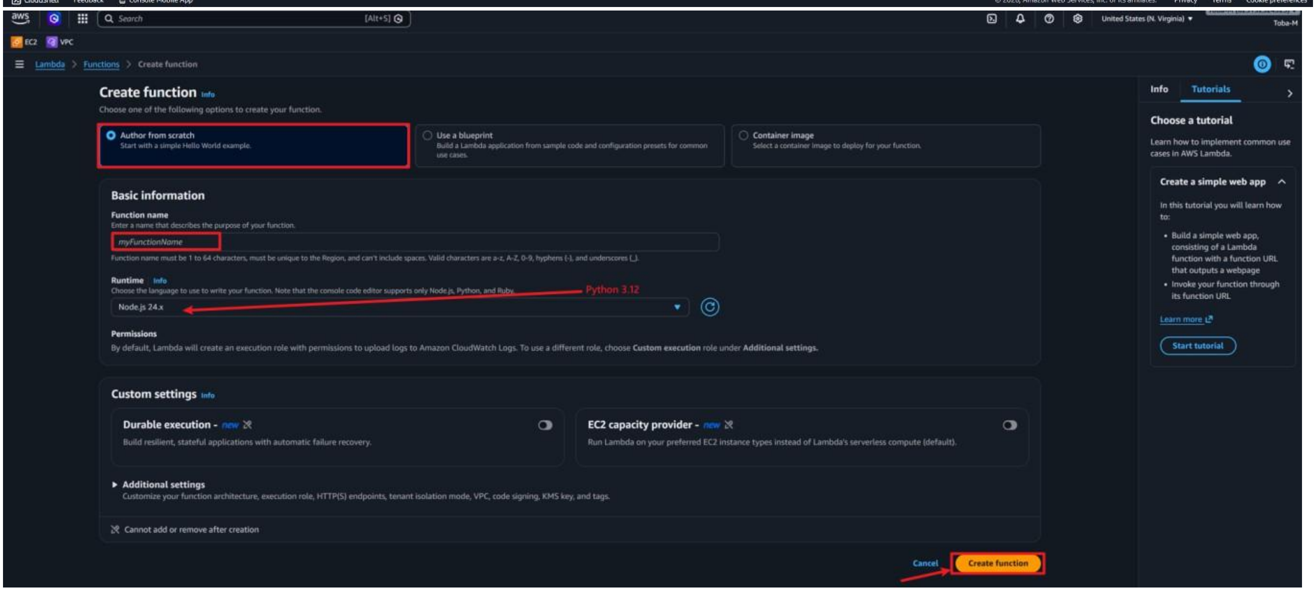
```
rotate-aws-to-huawei-source-secret
```

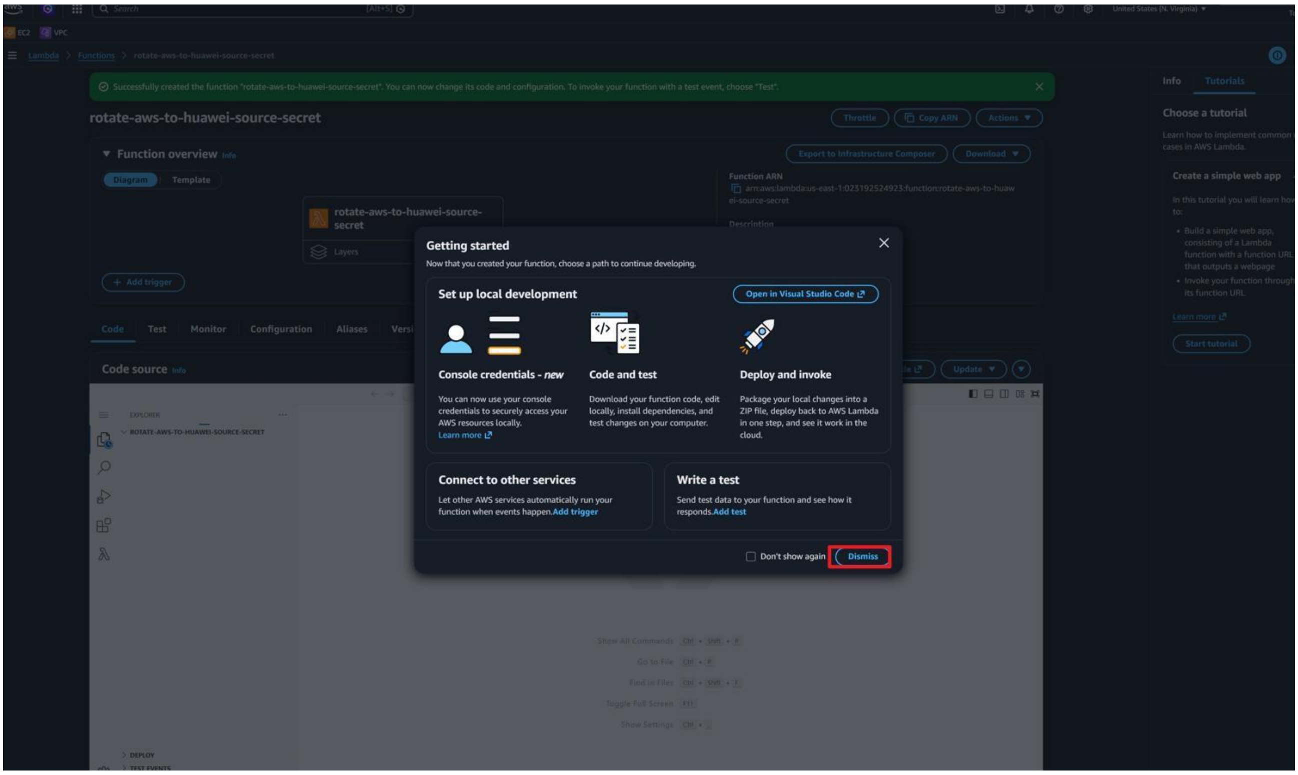
Set the runtime to **Python 3.12**.

Leave architecture as **x86_64** and leave permissions as **Create a new role with basic Lambda permissions**.

Click **Create function**.







Step 2 — Paste the rotation Lambda code

After the function is created, close any walkthrough popup.

Open the file `lambda_function.py` in the code editor.

Select all the default code and delete it.

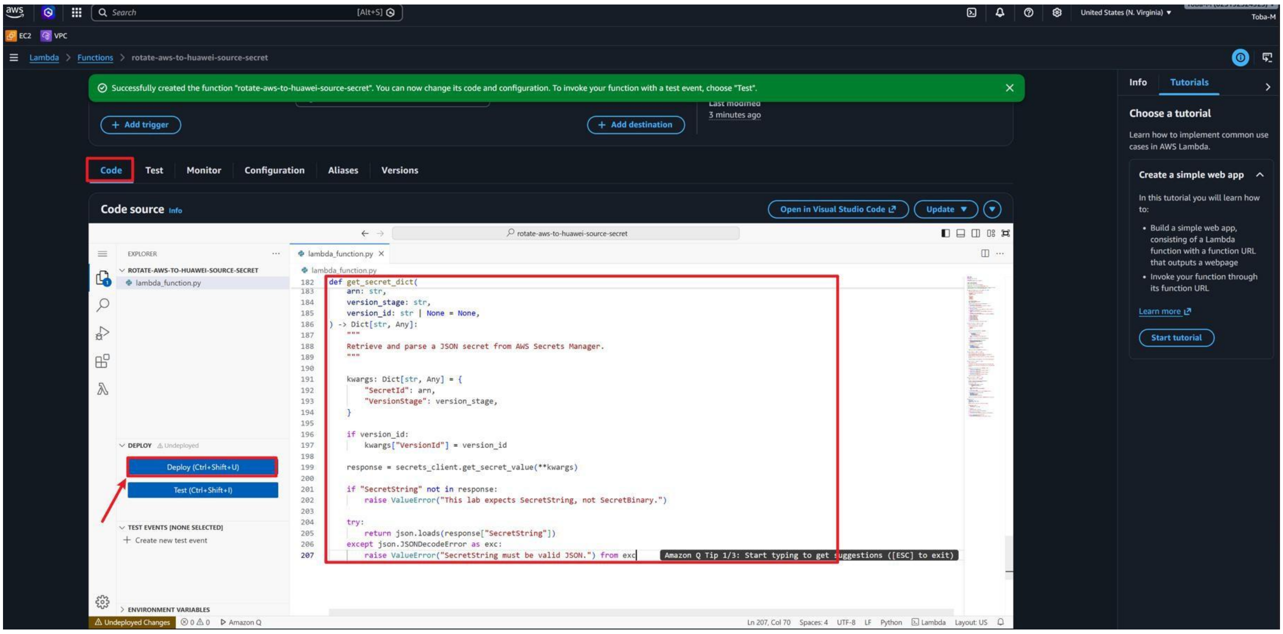
Paste the full rotation Lambda code below into the editor.

```
import json, logging, os
from typing import Any, Dict
import boto3

logger = logging.getLogger()
logger.setLevel(logging.INFO)
secrets_client = boto3.client("secretsmanager")
PASSWORD_LENGTH = int(os.environ.get("PASSWORD_LENGTH", "24"))
EXCLUDE_CHARACTERS = os.environ.get("EXCLUDE_CHARACTERS", '"'`\\/@')

def lambda_handler(event, context):
    arn   = event["SecretId"]
    token = event["ClientRequestToken"]
    step  = event["Step"]
    metadata = secrets_client.describe_secret(SecretId=arn)
    if not metadata.get("RotationEnabled"):
        raise ValueError(f"Secret {arn} is not enabled for rotation.")
    versions = metadata["VersionIdsToStages"]
    if token not in versions:
        raise ValueError(f"Secret version {token} has no stage for rotation.")
    if "AWSCURRENT" in versions[token]: return
    if "AWSPENDING" not in versions[token]:
        raise ValueError(f"Secret version {token} is not marked as AWSPENDING.")
    if   step == "createSecret": create_secret(arn, token)
    elif step == "setSecret":     set_secret(arn, token)
    elif step == "testSecret":    test_secret(arn, token)
    elif step == "finishSecret": finish_secret(arn, token)
    else: raise ValueError(f"Invalid rotation step: {step}")
```

```
# See full source in the lab repository for create_secret,
# set_secret, test_secret, finish_secret, and get_secret_dict.
```

NOTE — Full code
The abbreviated listing above shows the entry point and structure.
Paste the complete Python code from the lab brief (all helpers included) before clicking Deploy.

Click **Deploy**.

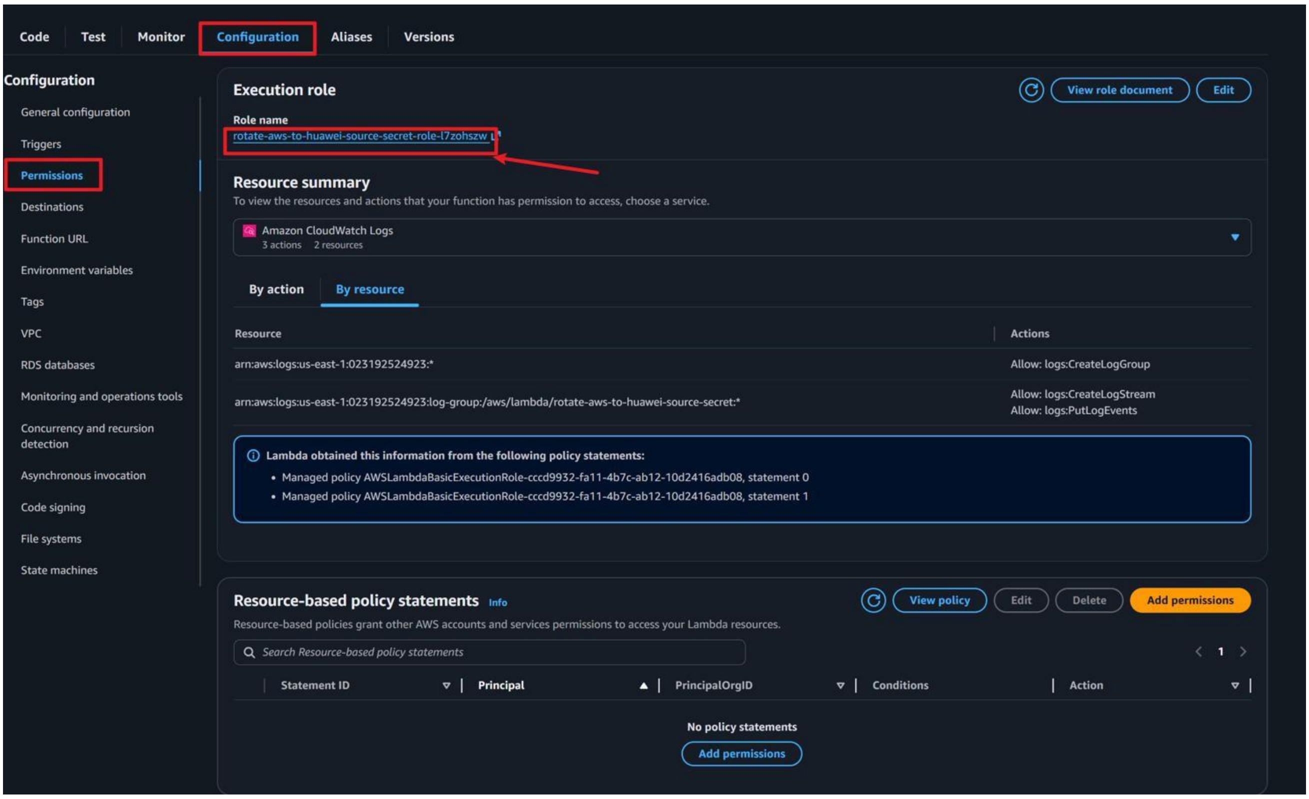


STAGE 3
GRANT PERMISSIONS TO THE ROTATION LAMBDA

Step 1 — Open the Lambda execution role

In the rotation Lambda, open **Configuration** → **Permissions**.

Click the execution role link to open it in IAM.



Step 2 — Create an inline policy

In IAM, click **Add permissions** → **Create inline policy**.

Open the **JSON** tab.

Paste this policy:

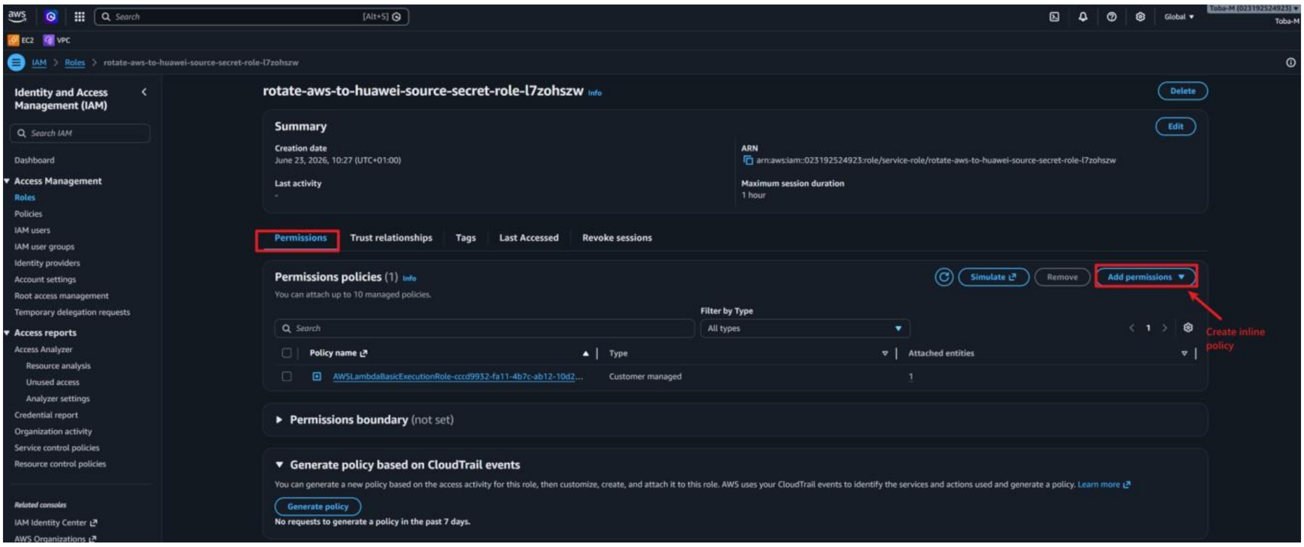
```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "AllowRotateAwsToHuaweiSourceSecret",
      "Effect": "Allow",
      "Action": [
        "secretsmanager:DescribeSecret",
        "secretsmanager:GetSecretValue",
        "secretsmanager:PutSecretValue",
        "secretsmanager:UpdateSecretVersionStage"
      ],
      "Resource": "arn:aws:secretsmanager:us-east-1:*:secret:lab/aws-to-huawei/source-secret-*"
    },
    {
      "Sid": "AllowGenerateRandomPassword",
      "Effect": "Allow",
      "Action": "secretsmanager:GetRandomPassword",
      "Resource": "*"
    }
```

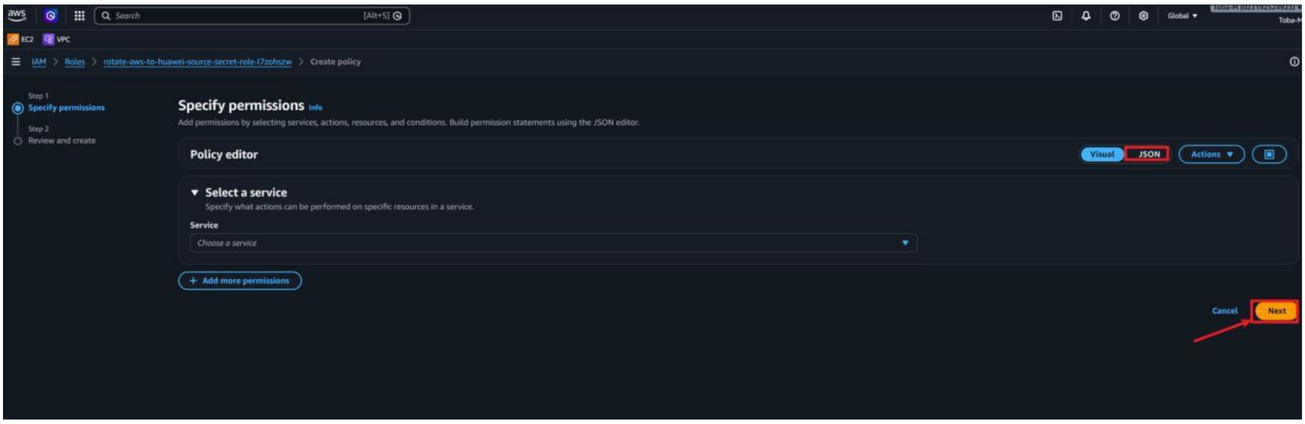
```
    ]
}
```

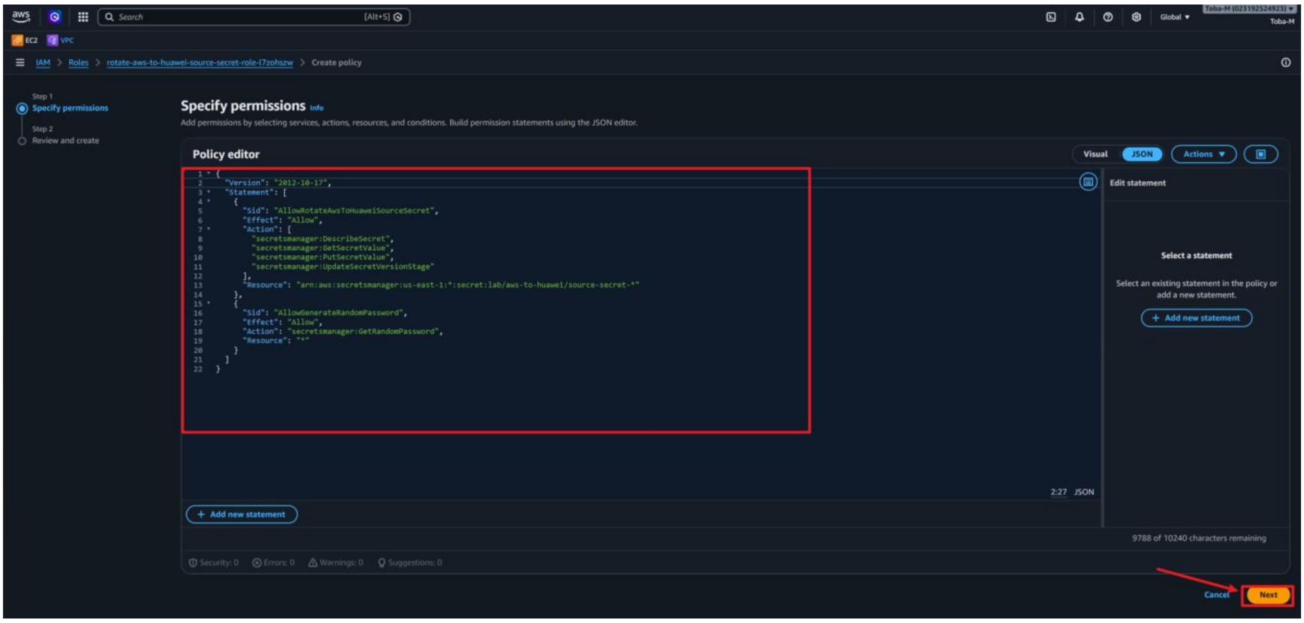
Click **Next**.

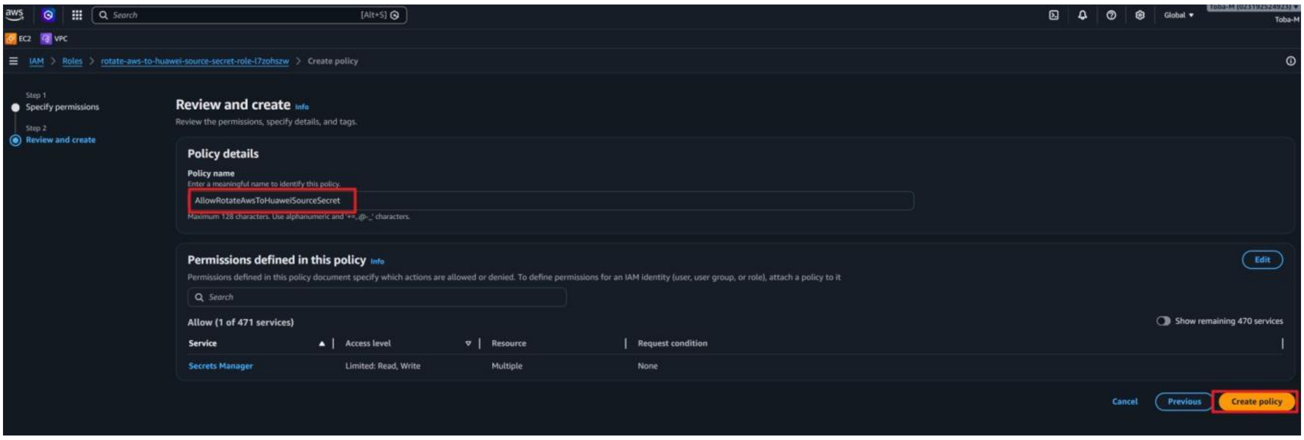
Set the policy name to `AllowRotateAwsToHuaweiSourceSecret`.

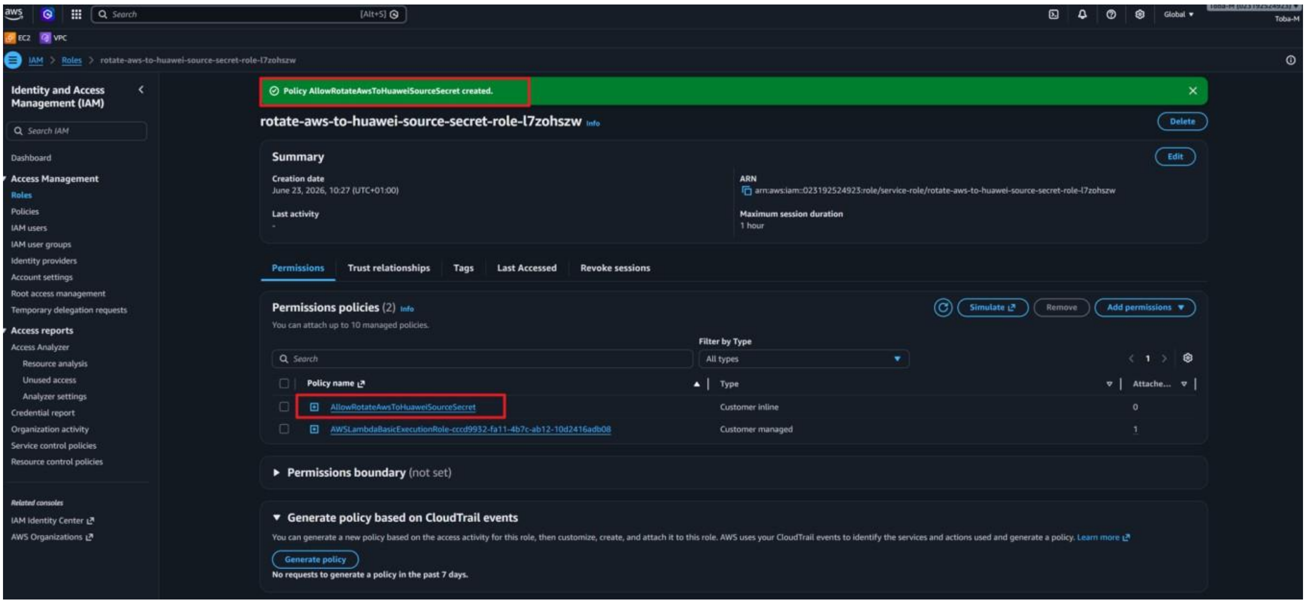
Click **Create policy**.











Step 3 — Allow Secrets Manager to invoke the rotation Lambda

Open **AWS CloudShell** from the top bar of the AWS console.

Run this command:

```
REGION="us-east-1"
SECRET_ID="lab/aws-to-huawei/source-secret"
FUNCTION_NAME="rotate-aws-to-huawei-source-secret"
ACCOUNT_ID=$(aws sts get-caller-identity --query Account --output text)
SECRET_ARN=$(aws secretsmanager describe-secret \
  --secret-id "$SECRET_ID" --query ARN --output text --region "$REGION")
aws lambda add-permission \
  --function-name "$FUNCTION_NAME" \
  --statement-id AllowSecretsManagerInvokeRotation \
  --action lambda:InvokeFunction \
  --principal secretsmanager.amazonaws.com \
  --source-account "$ACCOUNT_ID" \
  --source-arn "$SECRET_ARN" \
  --region "$REGION"
```

If this worked, you should see: a JSON object containing a `"Statement"` key in the CloudShell output.

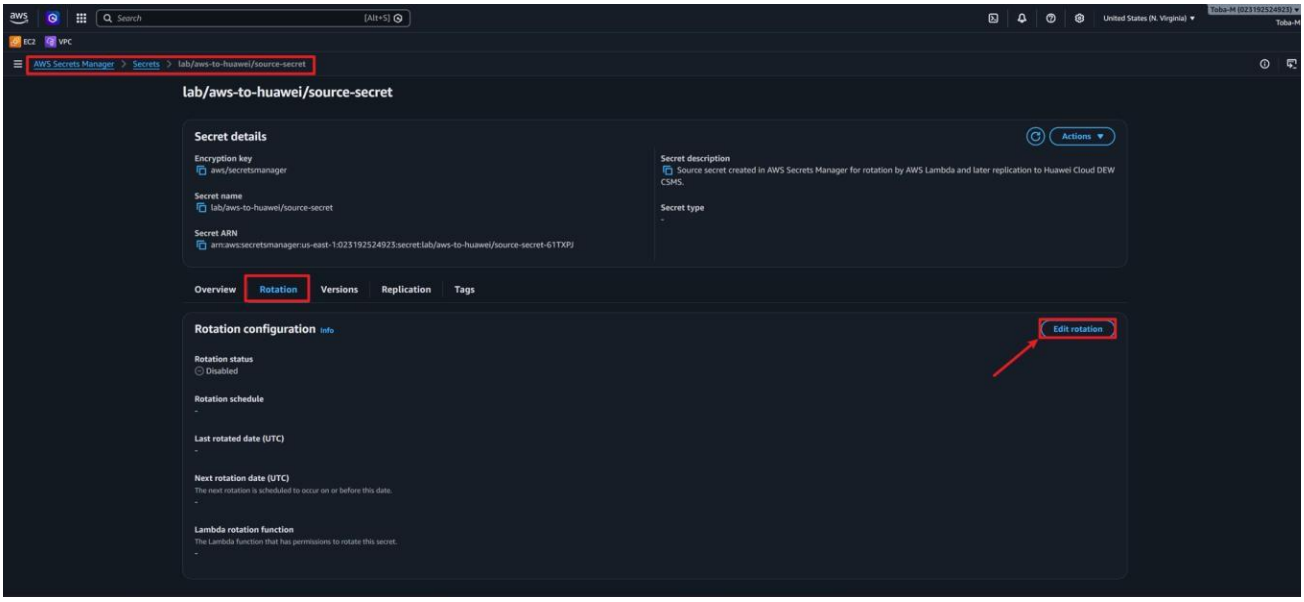
STAGE 4
ENABLE ROTATION ON THE SOURCE SECRET

Step 1 — Open the secret's rotation tab

Go to **AWS Secrets Manager** → `lab/aws-to-huawei/source-secret`.

Click the **Rotation** tab.

Click **Edit rotation**.



Step 2 — Configure rotation settings

Turn on **Automatic rotation**.

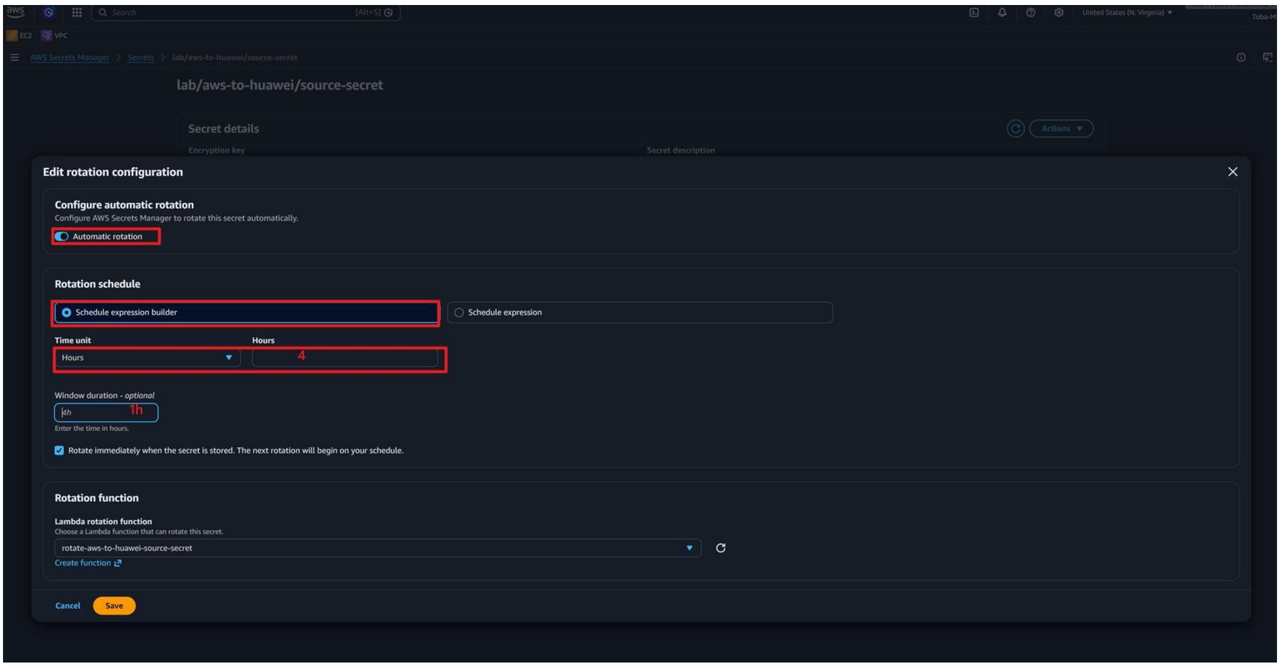
Set the schedule expression to `rate(4 hours)`.

Set the rotation window to `1h`.

Choose the rotation Lambda: `rotate-aws-to-huawei-source-secret`.

Enable **Rotate immediately when the secret is stored**.

Click **Save**.

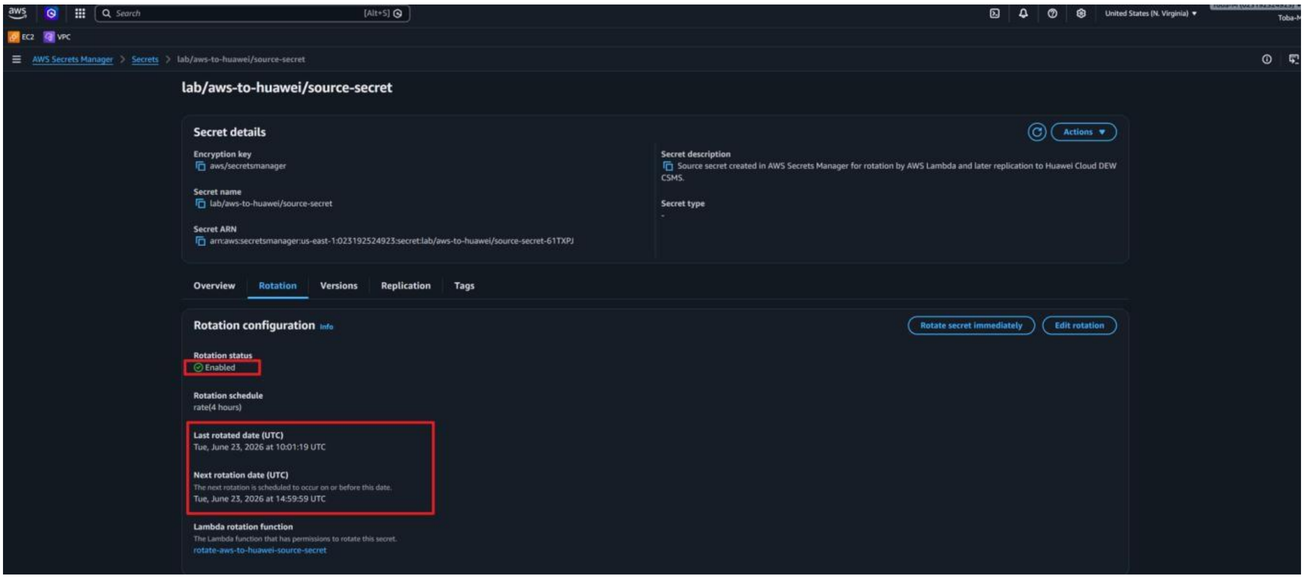


Step 3 — Confirm rotation succeeded

After saving, the rotation tab should show **Rotation status:** `Enabled`.

Open the Versions tab.

You should see two versions: one labelled `AWSCURRENT` and one labelled `AWSPREVIOUS`.



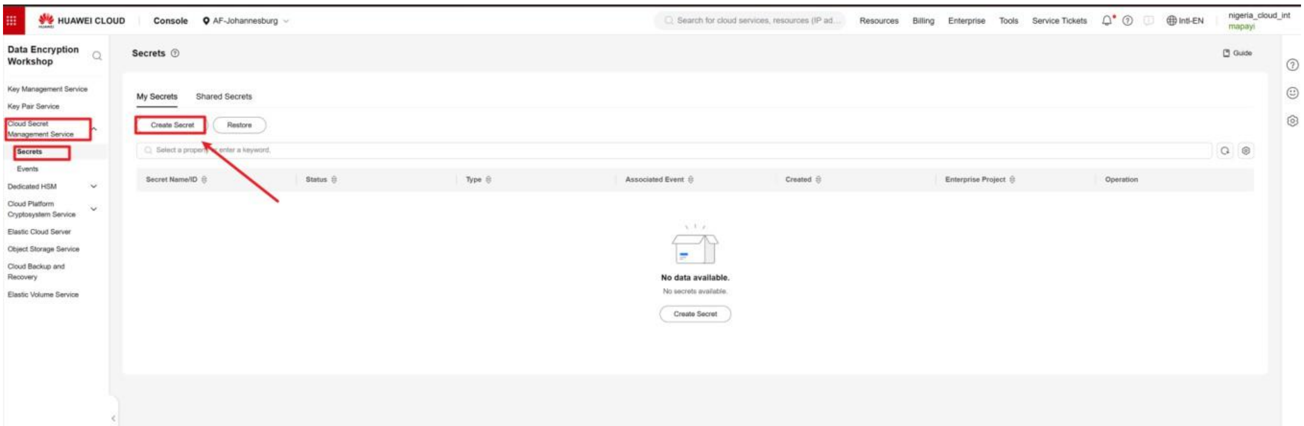
STAGE 5
CREATE THE HUAWEI CSMS TARGET SECRET

Step 1 — Open Huawei Cloud CSMS

Log in to the Huawei Cloud console.

Search for **Data Encryption Workshop** and open it.

In the left menu, click **Cloud Secret Management Service**.



Step 2 — Create the target secret

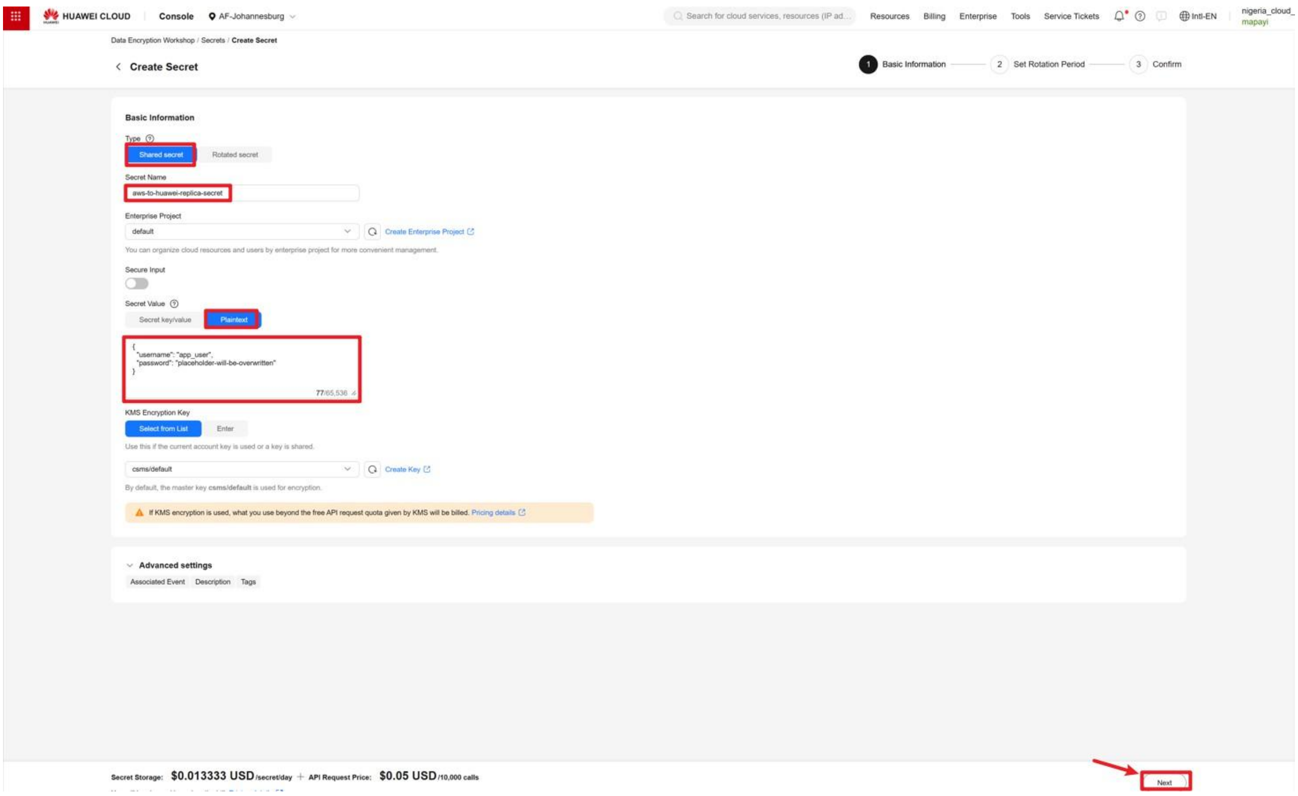
Click **Create Secret**.

Set the type to **Shared secret**.

Set the secret name to `aws-huawei_replica`.

Choose your lab enterprise project.

Leave the KMS encryption key as the default key.



Step 3 — Set a placeholder initial value

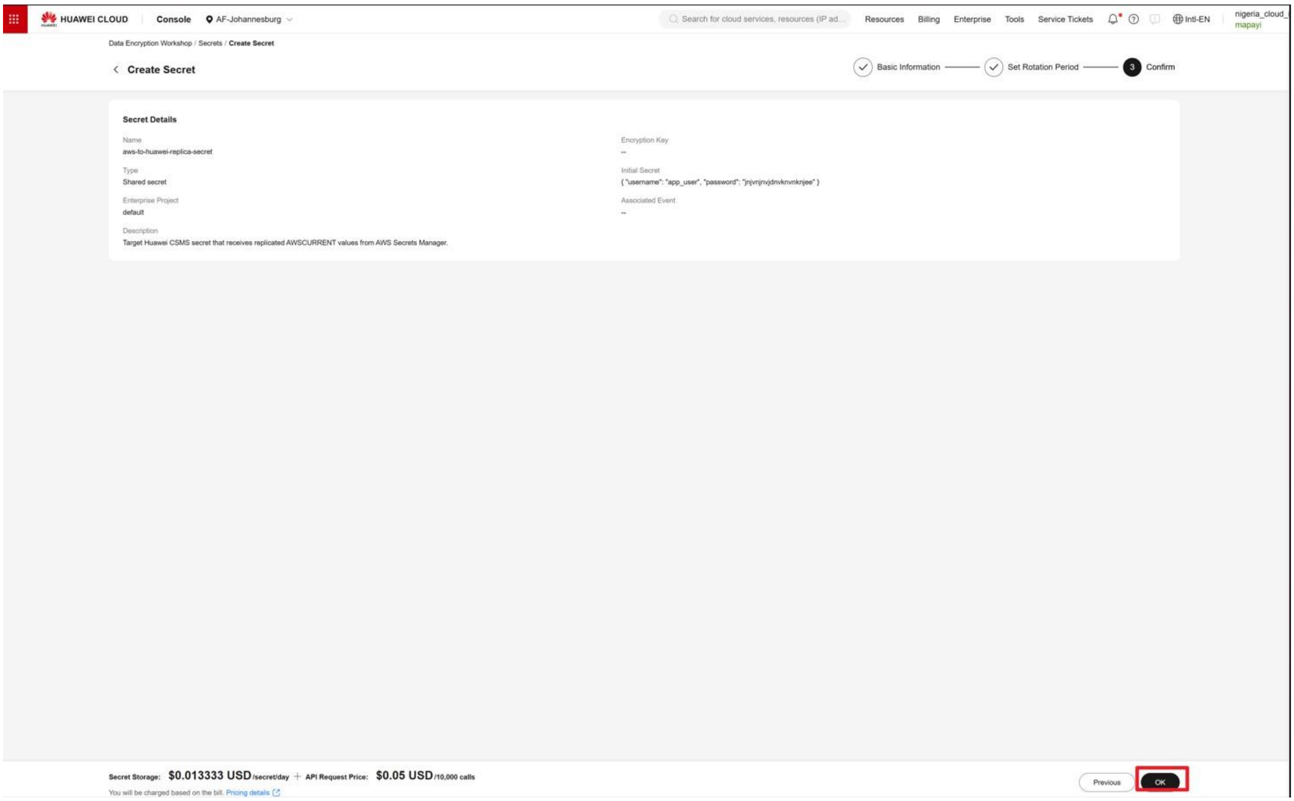
Enter a placeholder value — the sync Lambda will overwrite it later.

Use this placeholder JSON:

```
{
  "username": "app_user",
  "password": "placeholder-will-be-overwritten"
}
```

Do not enable Huawei-side rotation.

Click **Create**.



STAGE 6
CREATE THE SYNC LAMBDA

Step 1 — Create the function

Get your necessary keys

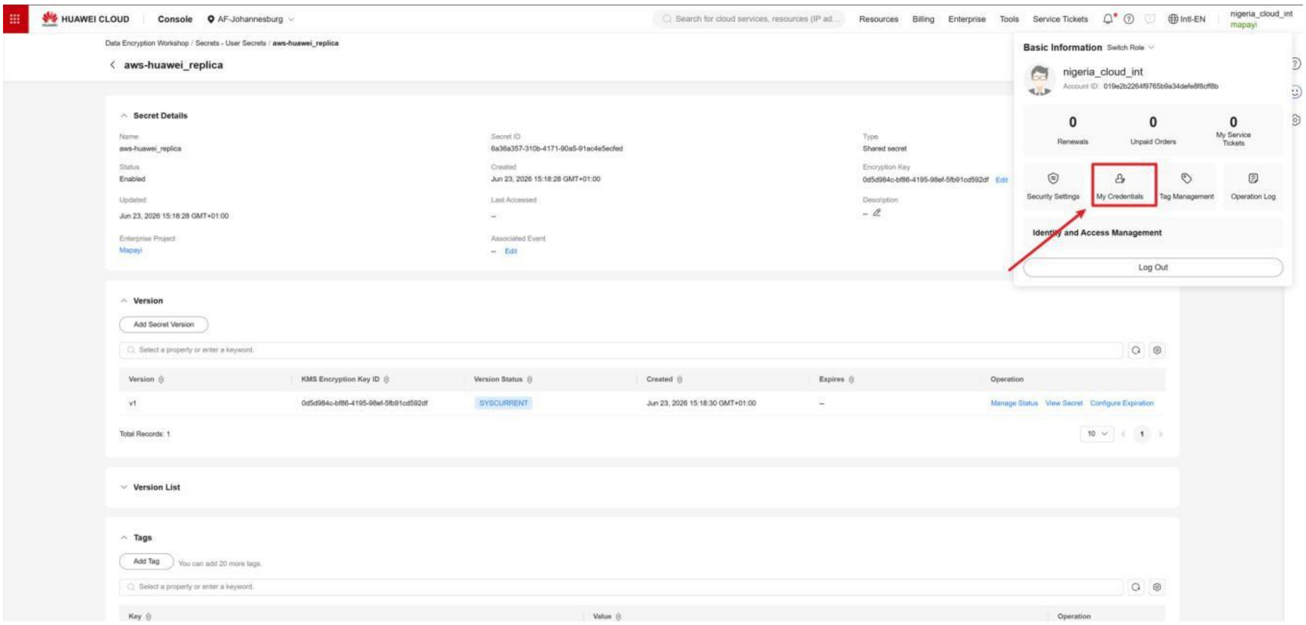
Go to **AWS Lambda** → **Create function**.

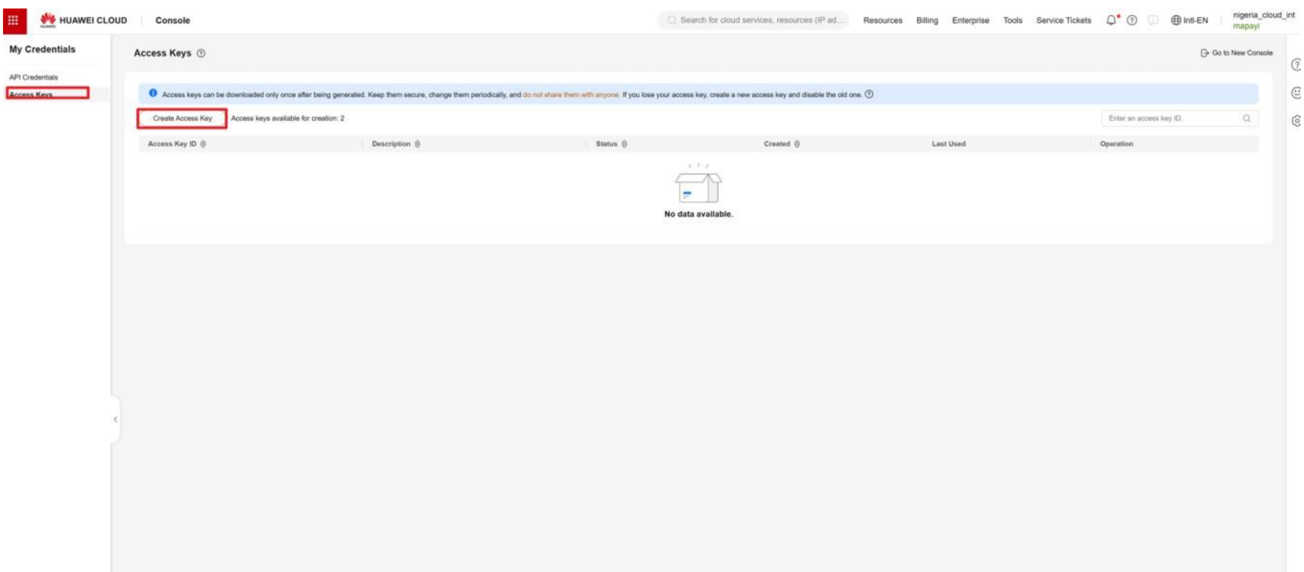
Choose **Author from scratch**.

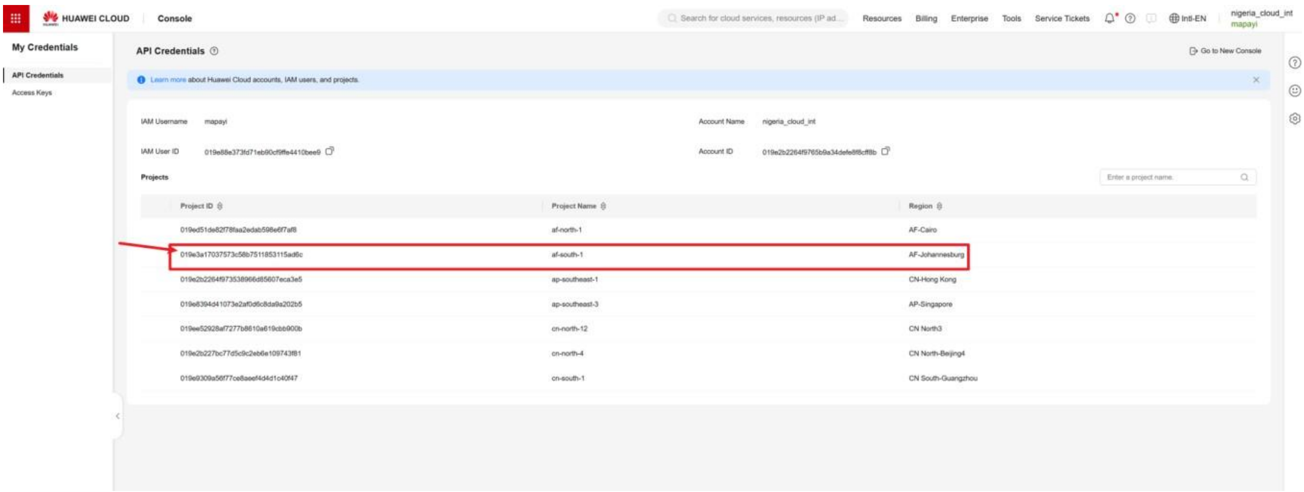
Set the function name to `sync-aws-secret-to-huawei-csms`.

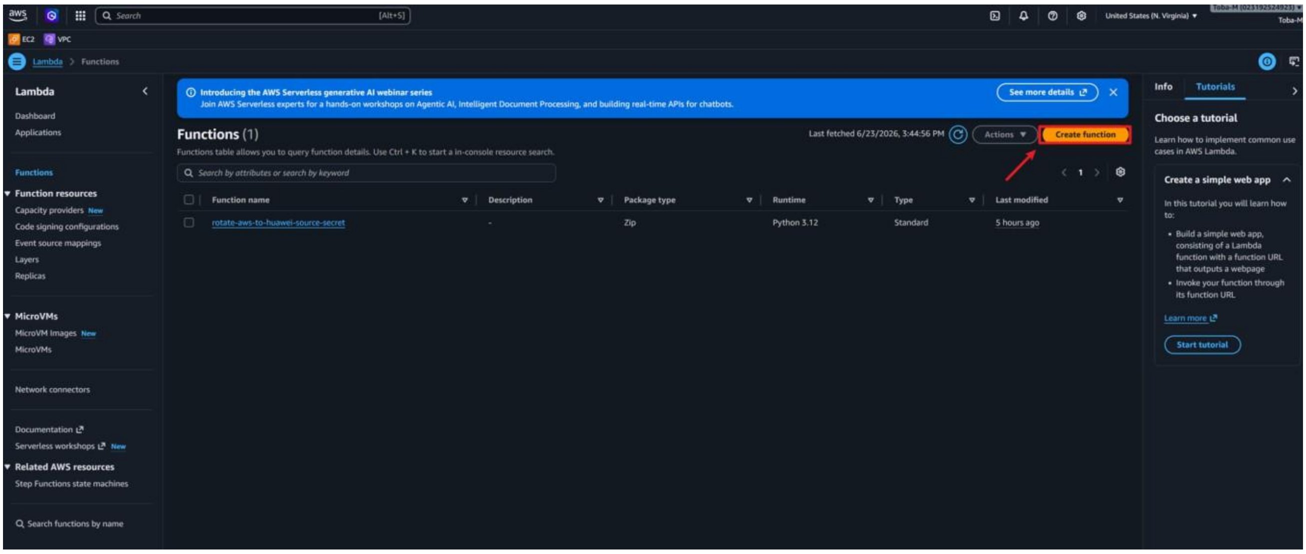
Set the runtime to **Python 3.12**.

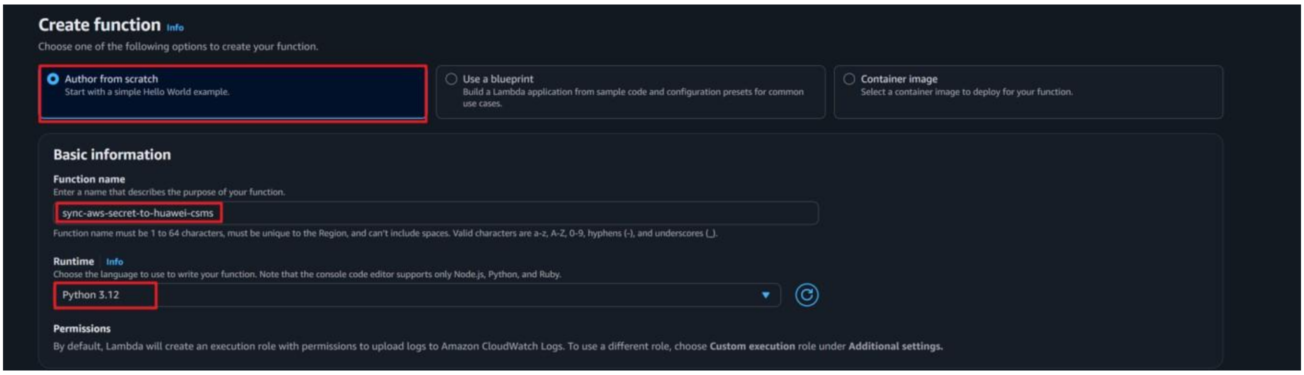
Click **Create function**.











Step 2 — Build and publish the Huawei SDK layer

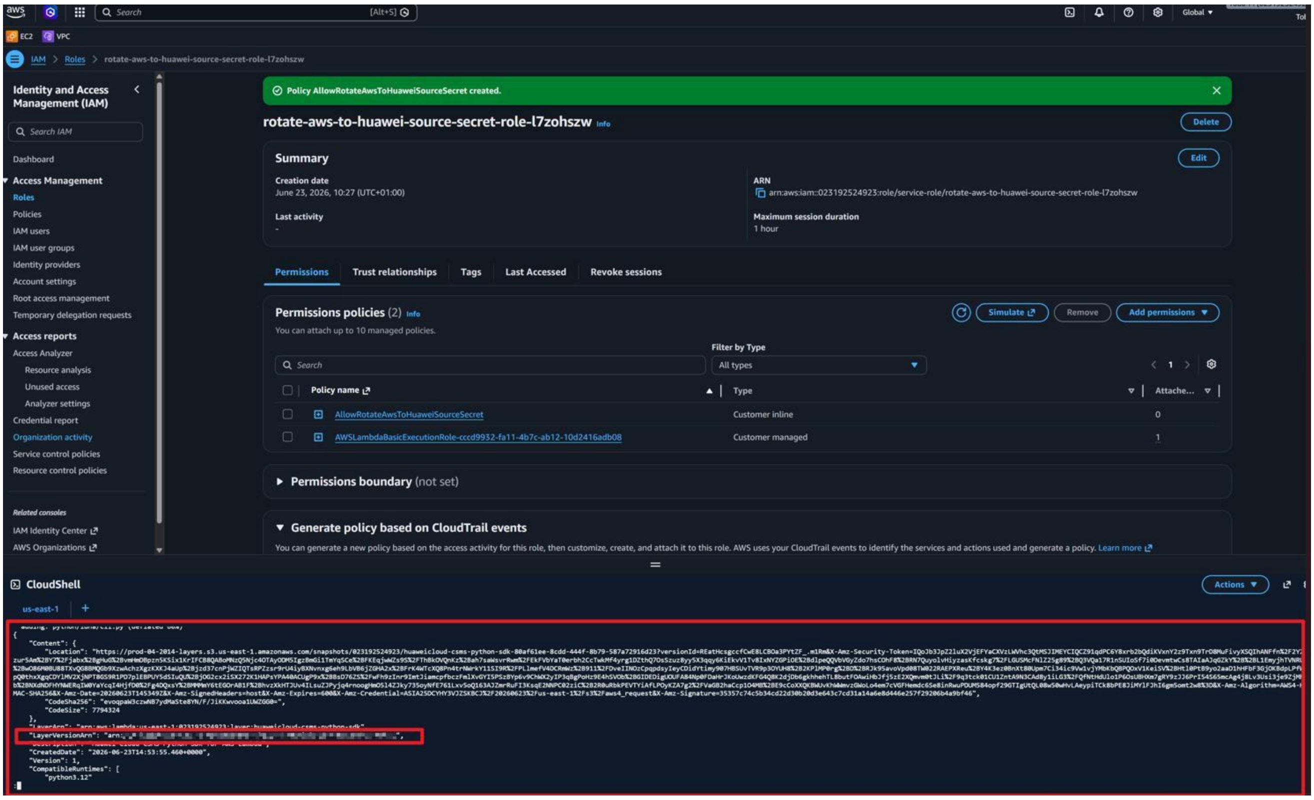
The sync Lambda needs the Huawei Python SDK installed as a Lambda Layer.

Open **AWS CloudShell** and run this command:

```
rm -rf huaweicloud-csms-layer huaweicloud-csms-layer.zip
mkdir -p huaweicloud-csms-layer/python
python3 -m pip install huaweicloudsdkcore huaweicloudsdkcsms \
  -t huaweicloud-csms-layer/python
cd huaweicloud-csms-layer
zip -r ../huaweicloud-csms-layer.zip python
cd ..
aws lambda publish-layer-version \
  --layer-name huaweicloud-csms-python-sdk \
  --description "Huawei Cloud CSMS Python SDK for AWS Lambda" \
  --zip-file fileb://huaweicloud-csms-layer.zip \
  --compatible-runtimes python3.12 \
```

```
  --region us-east-1
```

Copy the `LayerVersionArn` value from the output — you will need it in the next step.



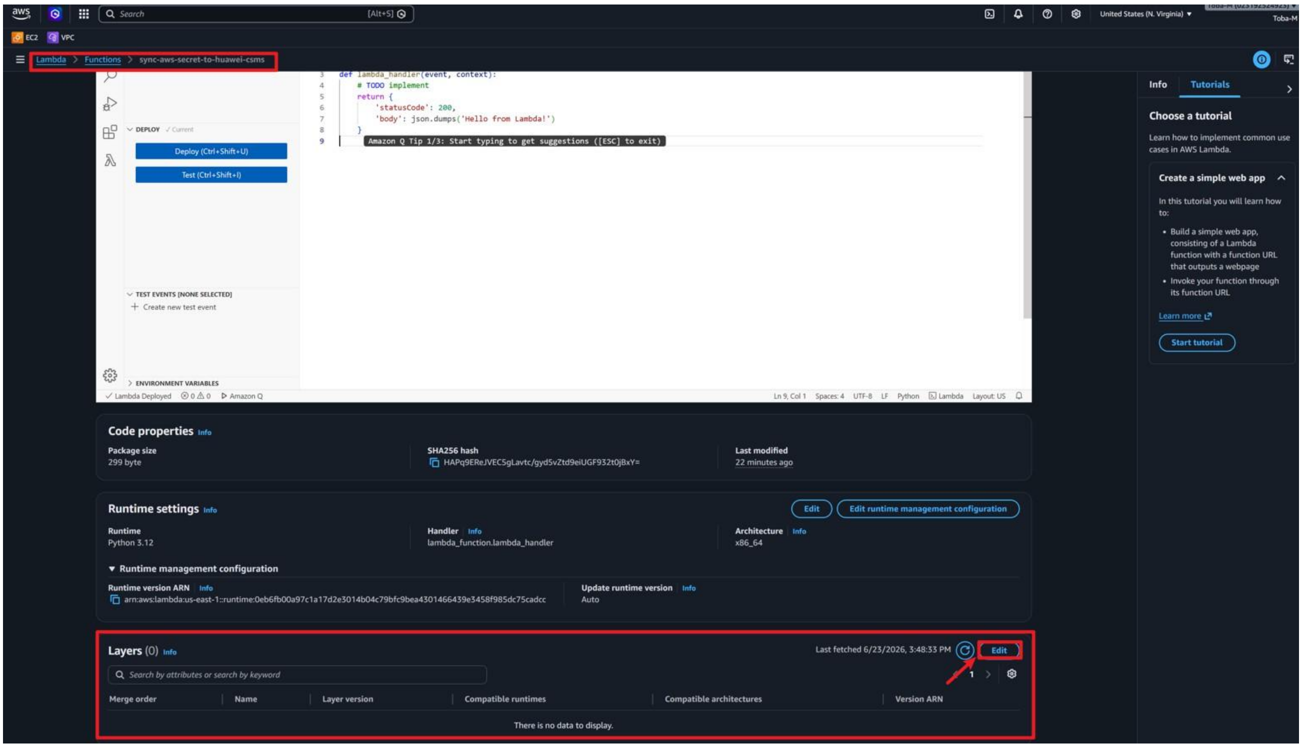
Step 3 — Attach the layer to the sync Lambda

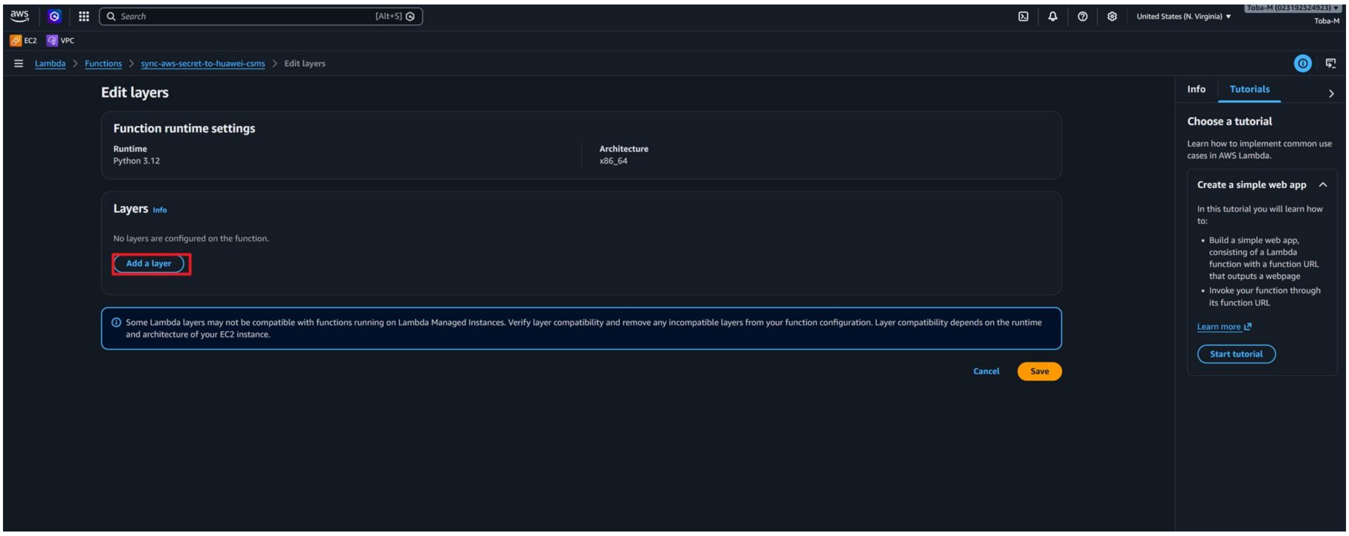
Go to **Lambda** → `sync-aws-secret-to-huawei-csms` → **Layers** → **Add a layer**.

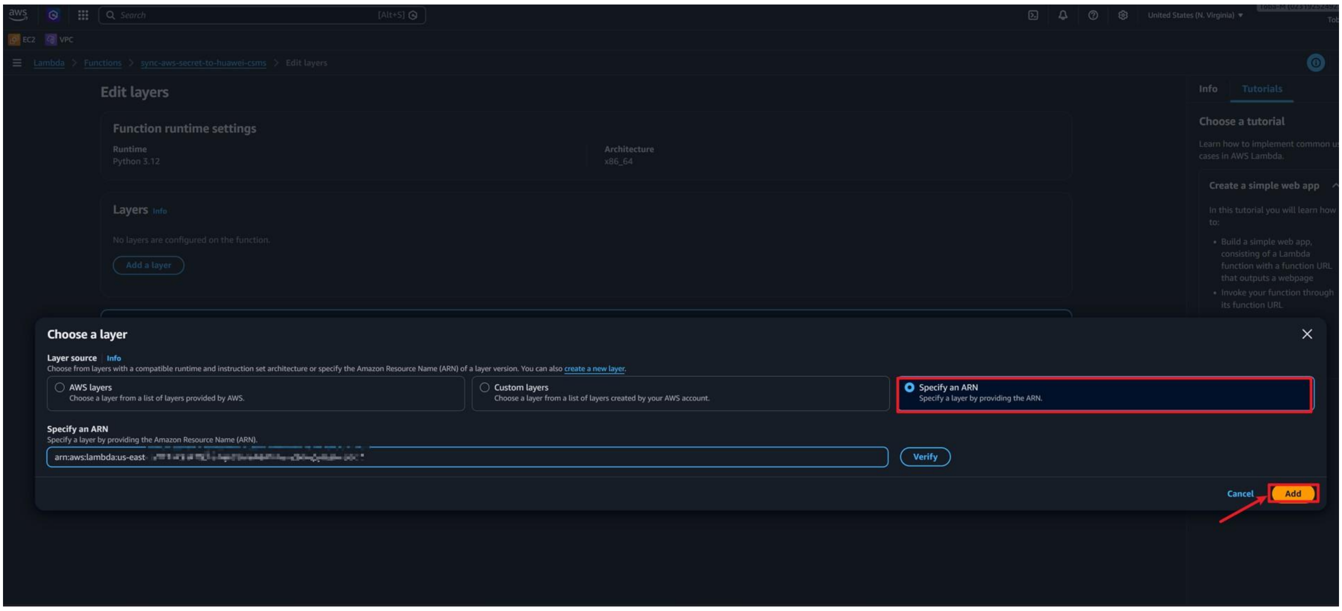
Choose **Specify an ARN**.

Paste the layer ARN you copied.

Click **Add**.







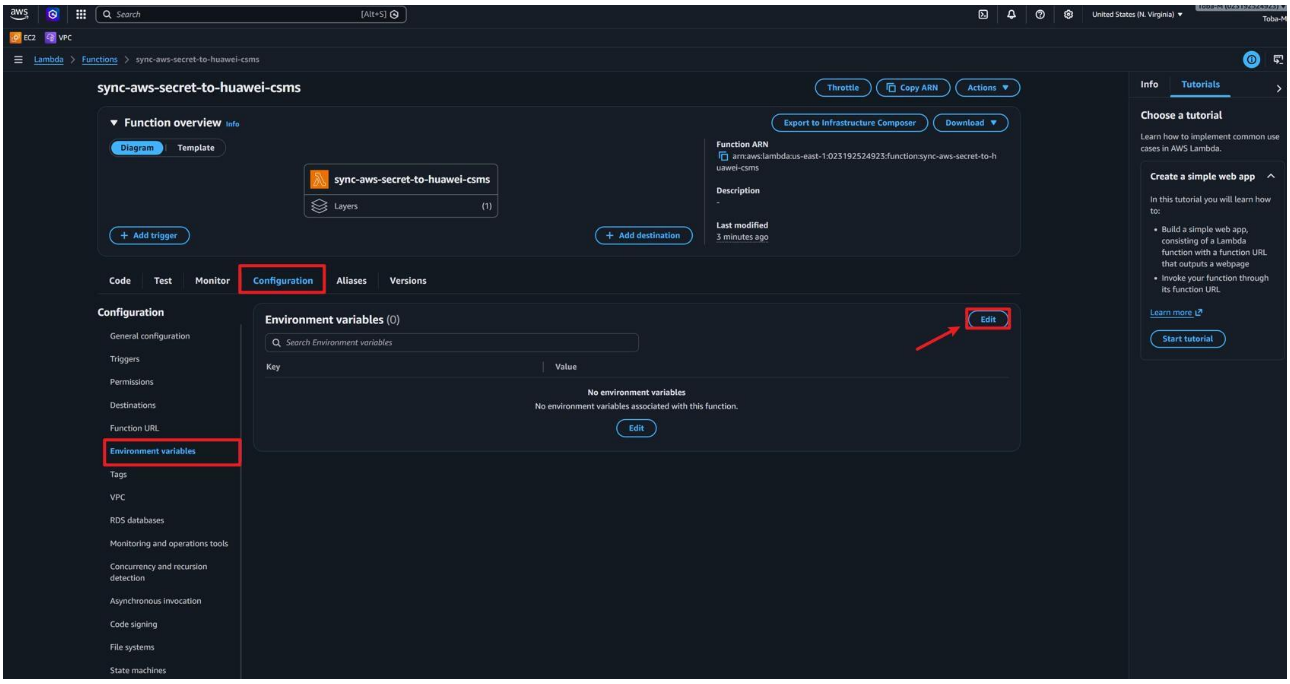
Step 4 — Add environment variables

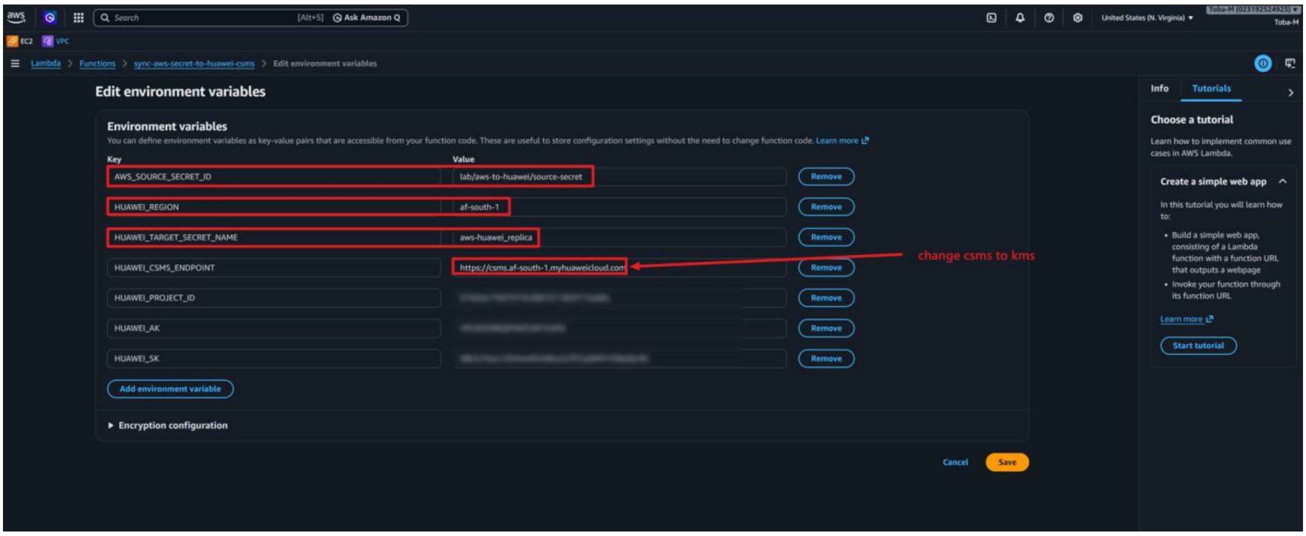
Go to **Configuration** → **Environment variables** → **Edit**.

Add these variables:

AWS_SOURCE_SECRET_ID	lab/aws-to-huawei/source-secret
HUAWEI_REGION	af-south-1
HUAWEI_TARGET_SECRET_NAME	aws-huawei_replica
HUAWEI_CSMS_ENDPOINT	https://kms.af-south-1.myhuaweicloud.com
HUAWEI_PROJECT_ID	<your Huawei AF-Johannesburg project ID>
HUAWEI_AK	<your Huawei access key>
HUAWEI_SK	<your Huawei secret access key>

WARNING — Correct endpoint
Set HUAWEI_CSMS_ENDPOINT to https://kms.af-south-1.myhuaweicloud.com.
Do NOT use https://csms.af-south-1.myhuaweicloud.com — that domain does not resolve and will cause a HostUnreachableException.





Step 5 — Grant the sync Lambda permission to read the AWS secret

Run this in CloudShell:

```
 ROLE_ARN=$(aws lambda get-function-configuration \
    --function-name sync-aws-secret-to-huawei-csms \
   --query Role --output text --region us-east-1)
ROLE_NAME=${ROLE_ARN##*/}
aws iam put-role-policy \
  --role-name "$ROLE_NAME" \
  --policy-name AllowSyncLambdaReadAwsSourceSecret \
  --policy-document '{"Version":"2012-10-17","Statement":[
    {"Sid":"AllowReadAwsSourceSecret","Effect":"Allow",
     "Action":["secretsmanager:GetSecretValue","secretsmanager:DescribeSecret"],
     "Resource":"arn:aws:secretsmanager:us-east-1:*:secret:lab/aws-to-huawei/source-
secret-*"}
  ]}'
```

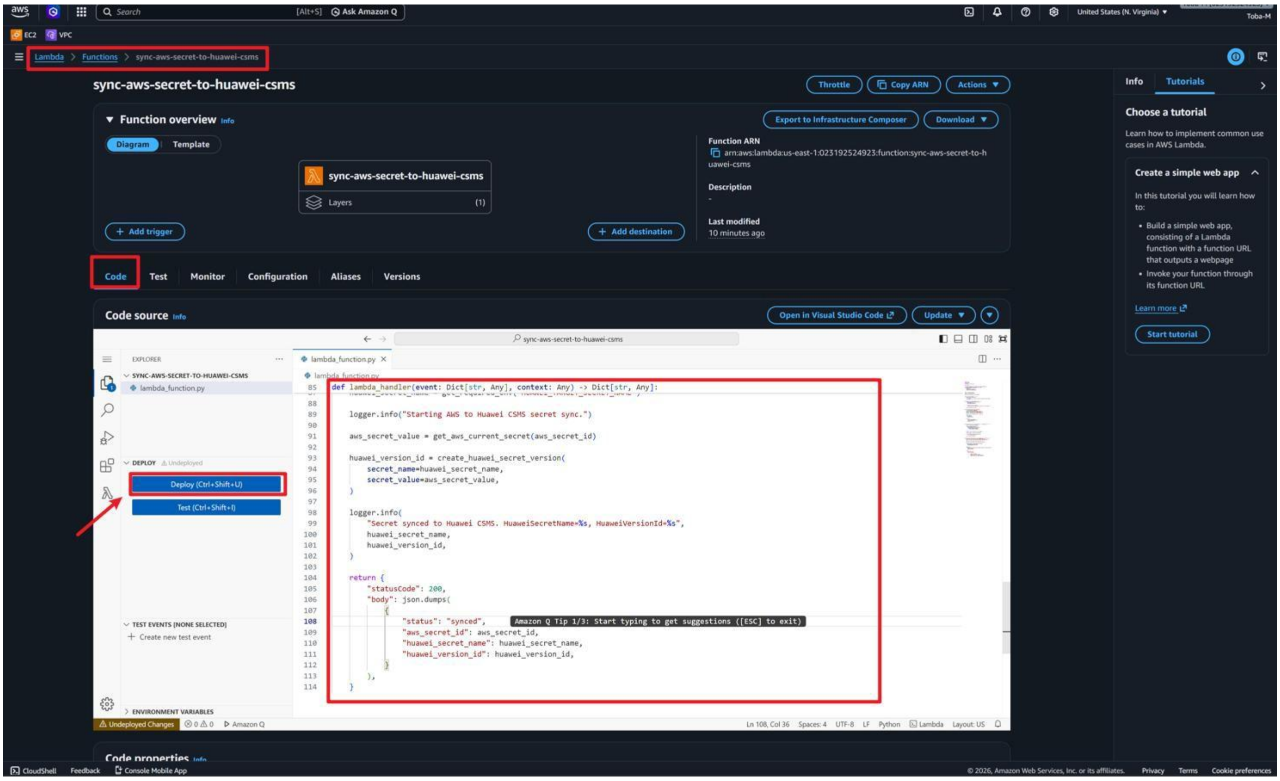
Step 6 — Paste the sync Lambda code

Open **Code** → `lambda_function.py`.

Delete the default code and paste the full sync Lambda code.

```
import json, logging, os
from typing import Any, Dict
import boto3
from huaweicloudsdkcore.auth.credentials import BasicCredentials
from huaweicloudsdkcore.region.region import Region
from huaweicloudsdkcsms.v1 import (CsmsClient,
    CreateSecretVersionRequest, CreateSecretVersionRequestBody)

logger = logging.getLogger()
logger.setLevel(logging.INFO)
aws_secrets = boto3.client("secretsmanager")

def get_required_env(name):
    value = os.environ.get(name)
    if not value: raise RuntimeError(f"Missing env var: {name}")
    return value

def lambda_handler(event, context):
    aws_secret_id      = get_required_env("AWS_SOURCE_SECRET_ID")
    huawei_secret_name = get_required_env("HUAWEI_TARGET_SECRET_NAME")
    logger.info("Starting AWS to Huawei CSMS secret sync.")
    response = aws_secrets.get_secret_value(
        SecretId=aws_secret_id, VersionStage="AWSCURRENT")
    secret_value = response["SecretString"]
    credentials = BasicCredentials(
        get_required_env("HUAWEI_AK"),
        get_required_env("HUAWEI_SK"),
        get_required_env("HUAWEI_PROJECT_ID"))
    region = Region(id=get_required_env("HUAWEI_REGION"),
                    endpoint=get_required_env("HUAWEI_CSMS_ENDPOINT"))
    client = (CsmsClient.new_builder()
        .with_credentials(credentials).with_region(region).build())
    request = CreateSecretVersionRequest(
        secret_name=huawei_secret_name,
        body=CreateSecretVersionRequestBody(
            secret_string=secret_value,
            version_stages=["SYSCURRENT"]))
    response = client.create_secret_version(request)
    version_id = response.version_metadata.id
    logger.info("Synced. HuaweiVersionId=%s", version_id)
    return {"statusCode": 200}
```

Click **Deploy**.



Step 7 — Increase the timeout

Go to **Configuration** → **General configuration** → **Edit**.

Set the timeout to `30 seconds`.

Click **Save**.

STAGE 7
TEST THE SYNC LAMBDA MANUALLY

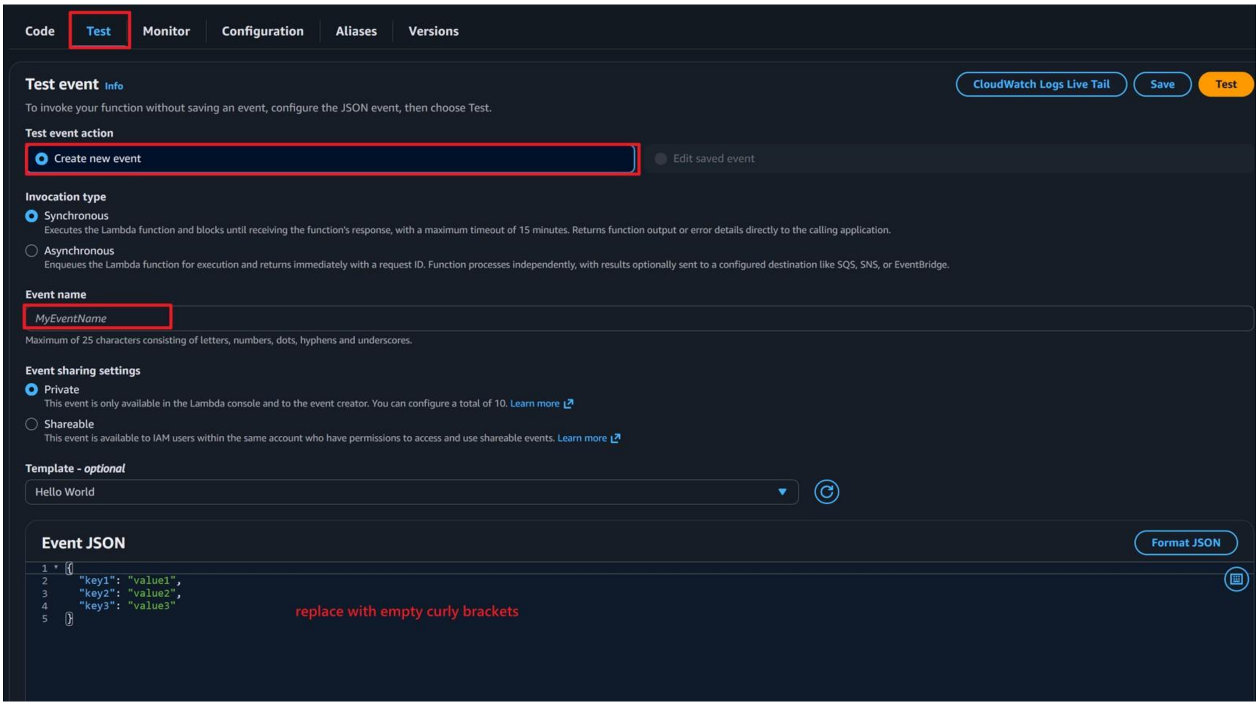
Step 1 — Create a test event

Open the **Test** tab in the sync Lambda.

Set the event name to `manual-sync-test`.

Set the event JSON to an empty object:

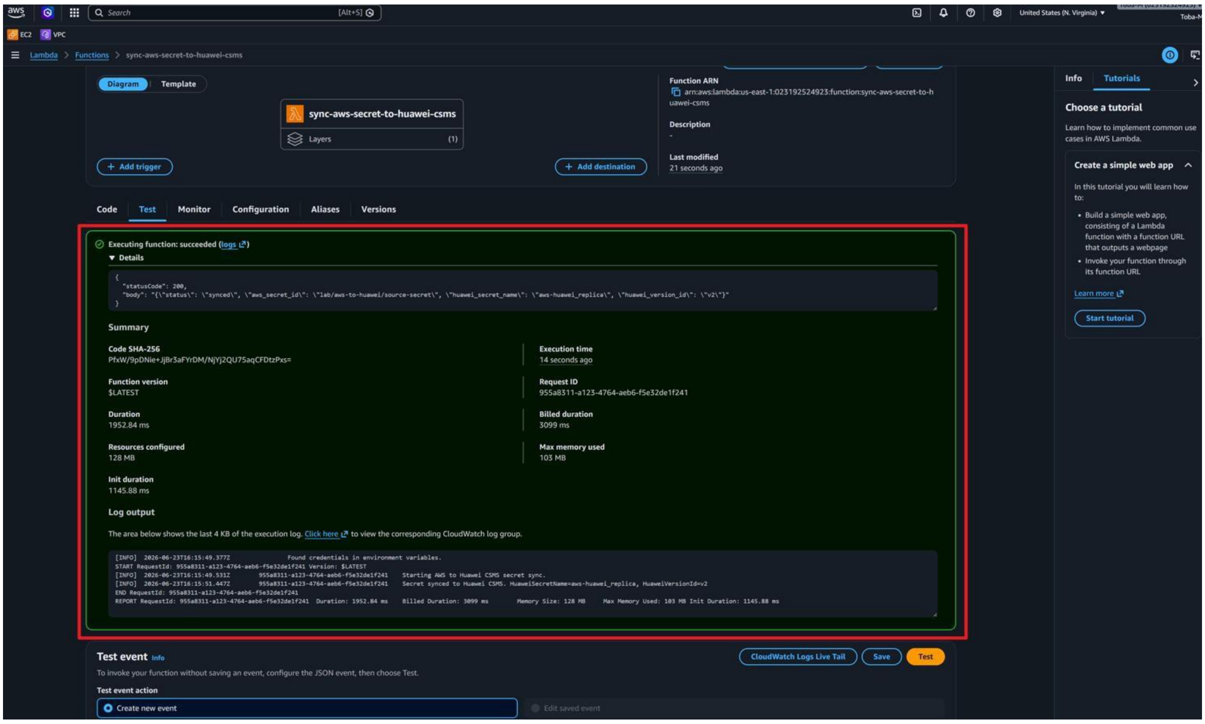
```
{}
```



Step 2 — Run the test

Click **Test**.

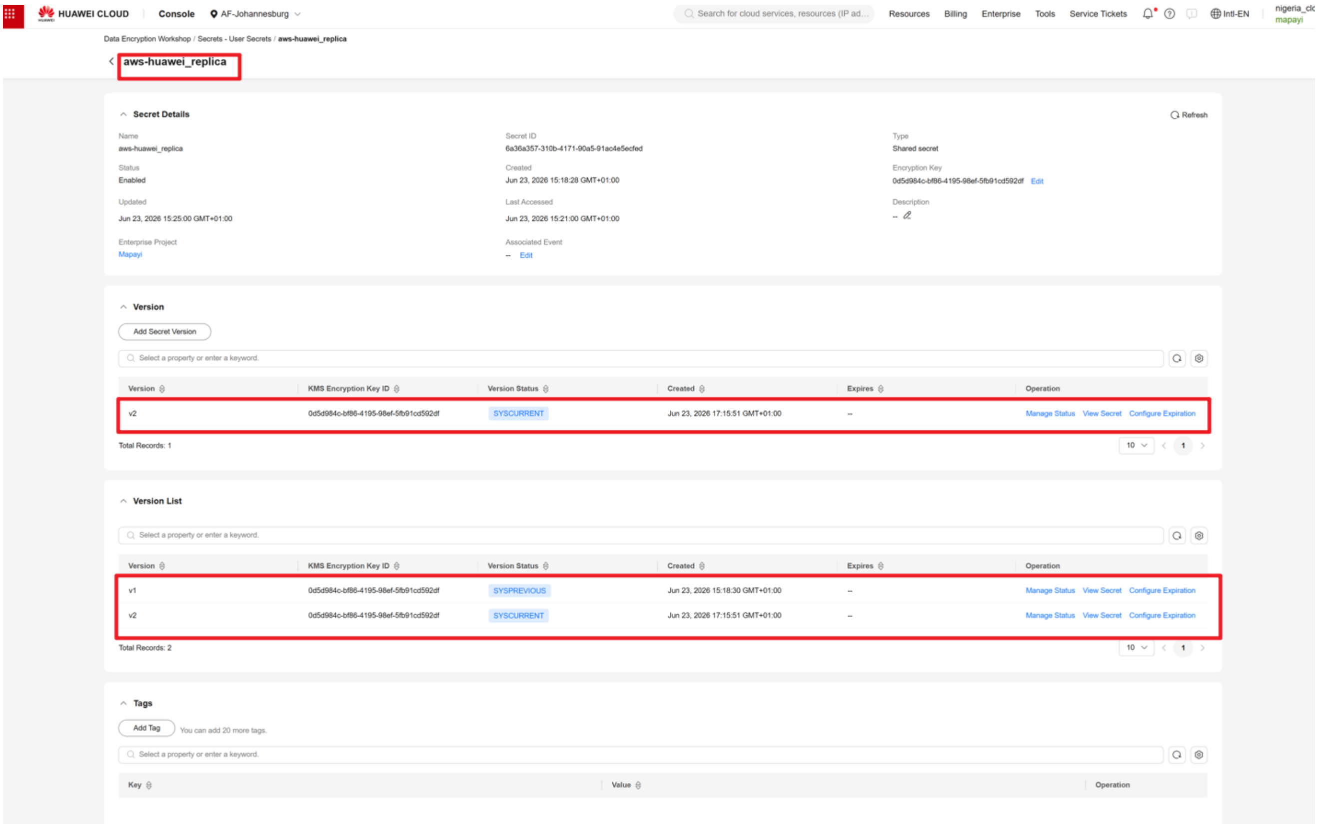
If this worked, you should see: a `statusCode: 200` response in the execution result.



Step 3 — Confirm the new version in Huawei CSMS

Open the Huawei Cloud console → **DEW** → **CSMS** → `aws-huawei_replica`.

You should see a new version with the stage `SYSCURRENT`.



WARNING — Wrong endpoint error

If the test fails with HostUnreachableException and "Failed to resolve csms.af-south-1.myhuaweicloud.com", go to Configuration → Environment variables and change HUAWEI_CSMS_ENDPOINT to https://kms.af-south-1.myhuaweicloud.com.

Run the test again after saving.

STAGE 8
CREATE THE EVENTBRIDGE RULE

Step 1 — Open EventBridge

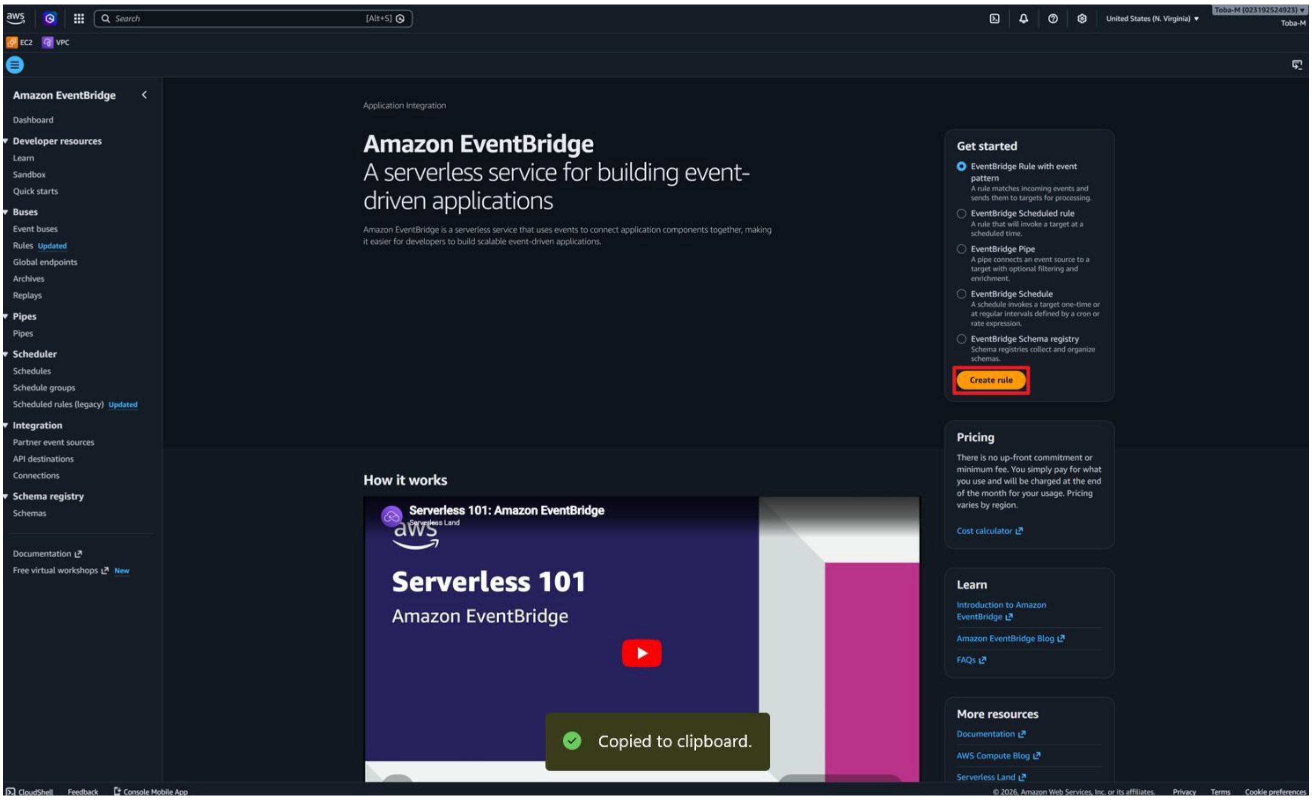
Go to **Amazon EventBridge** → **Rules** → **Create rule**.

Choose **Advanced builder**.

Set the rule name to `sync-huawei-after-aws-secret-rotation`.

Set the description to: *Invokes the Huawei CSMS sync Lambda after AWS Secrets Manager rotation succeeds.*

Set the event bus to **default**.

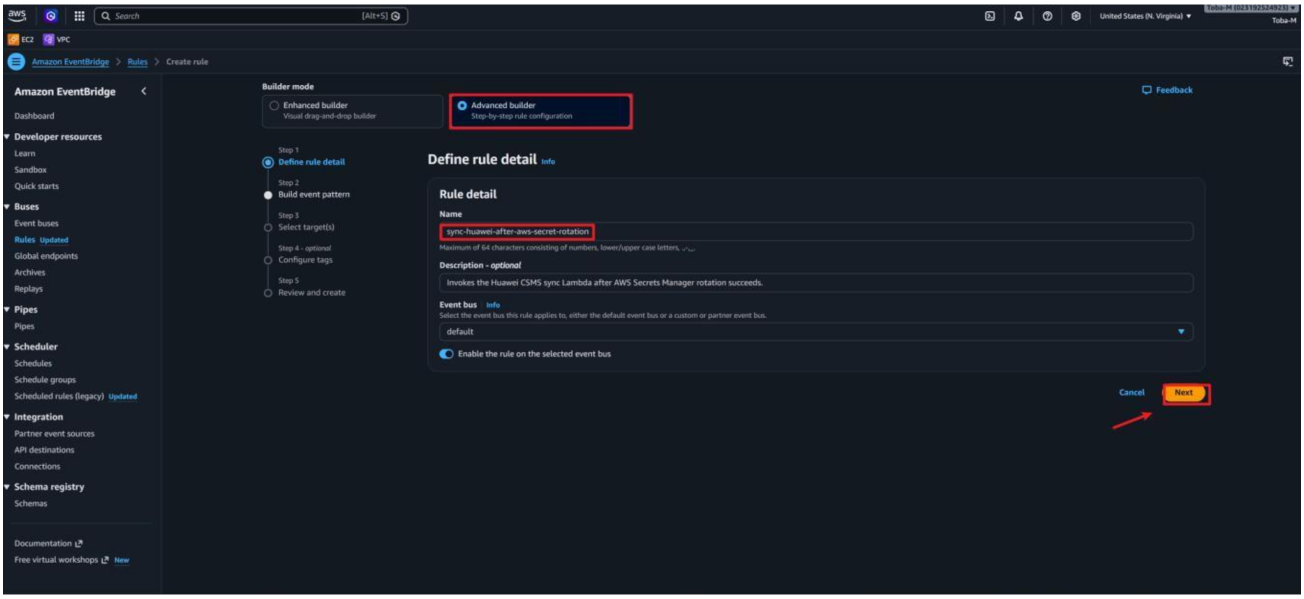


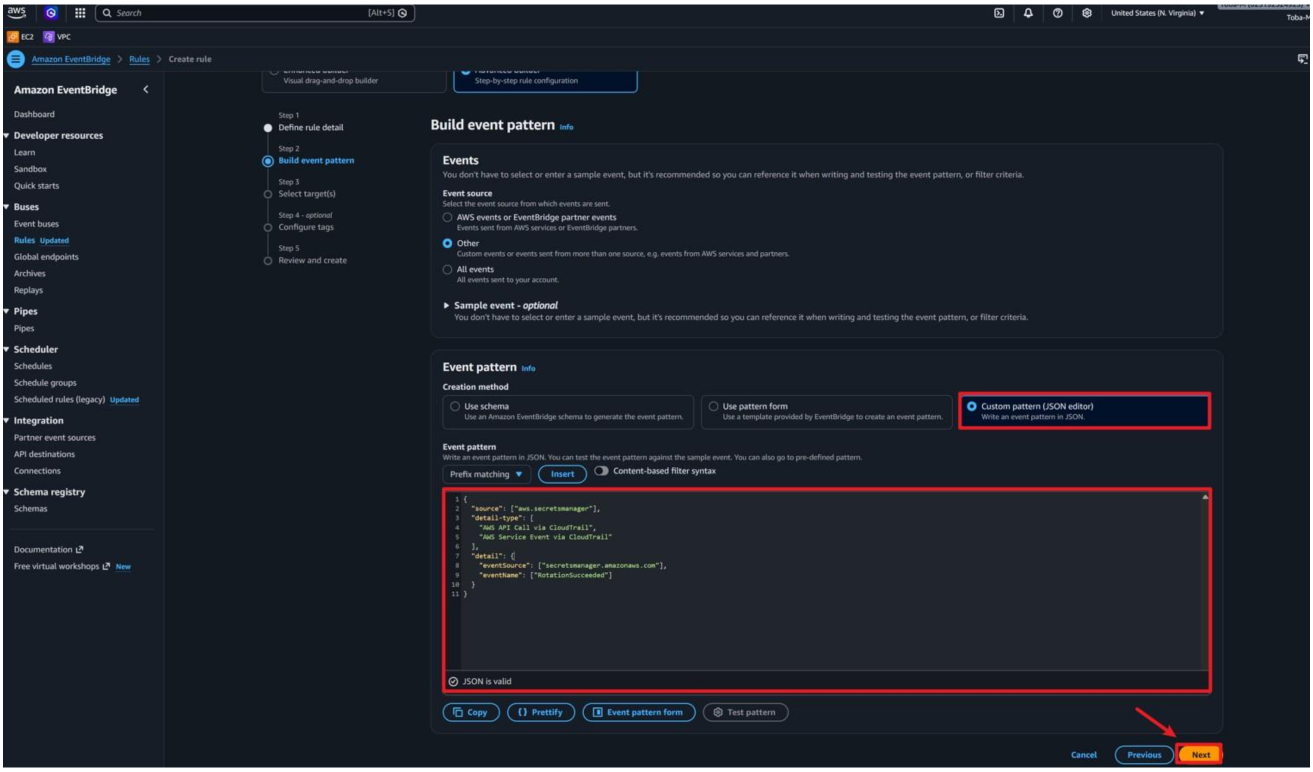
Step 2 — Set the event pattern

Paste this event pattern into the event pattern editor:

```
{
  "source": ["aws.secretsmanager"],
  "detail-type": ["AWS API Call via CloudTrail"],
  "detail": {
    "eventSource": ["secretsmanager.amazonaws.com"],
    "eventName": ["UpdateSecretVersionStage"]
  }
}
```

> **NOTE — Why UpdateSecretVersionStage?**
> Secrets Manager fires this API call when it promotes a new secret version to AWSCURRENT.
> This is the reliable signal that rotation is complete and a new value is ready to replicate.





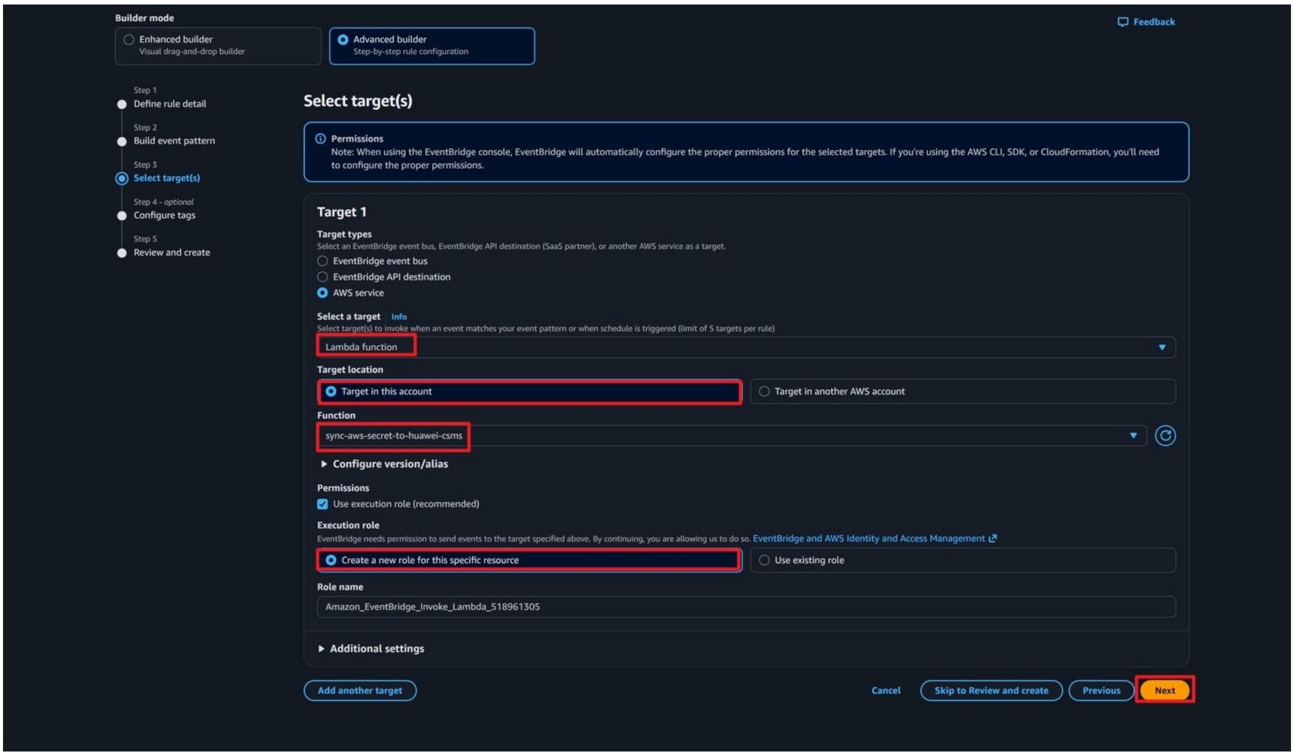
Step 3 — Set the target

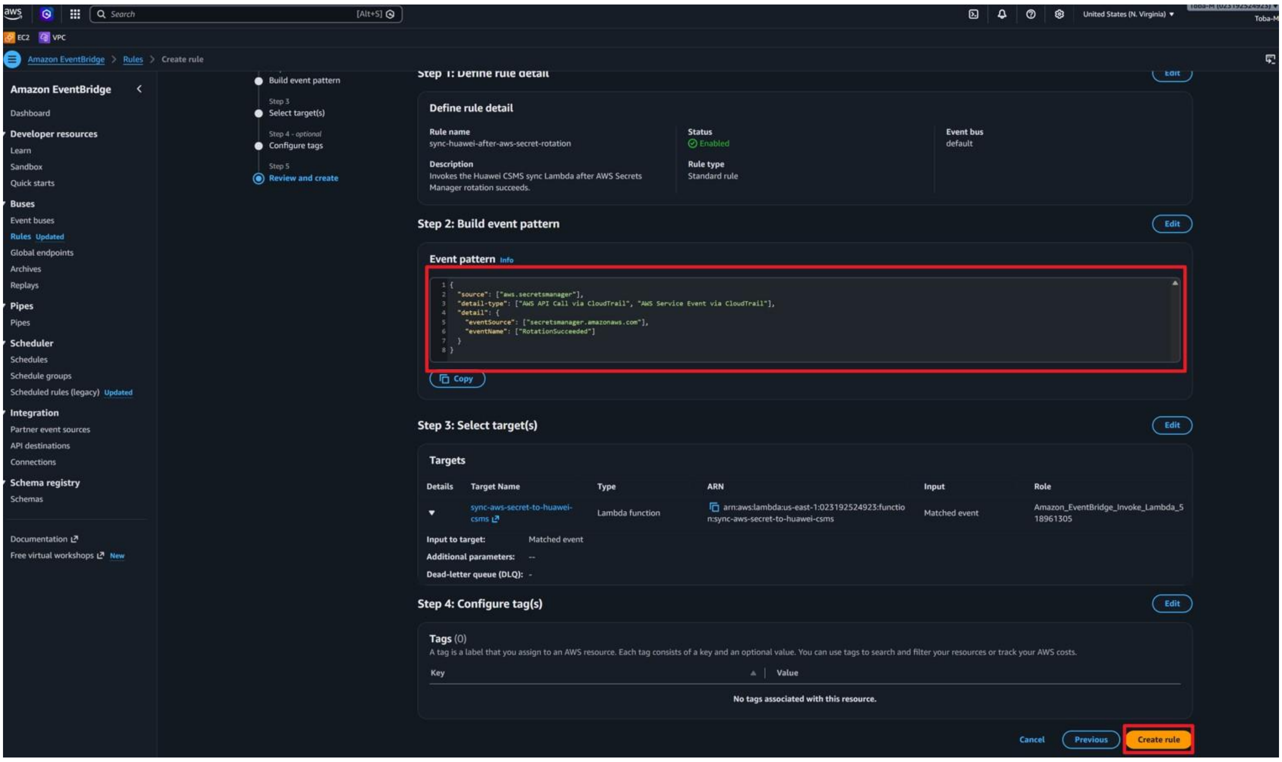
Set target type to **AWS service**.

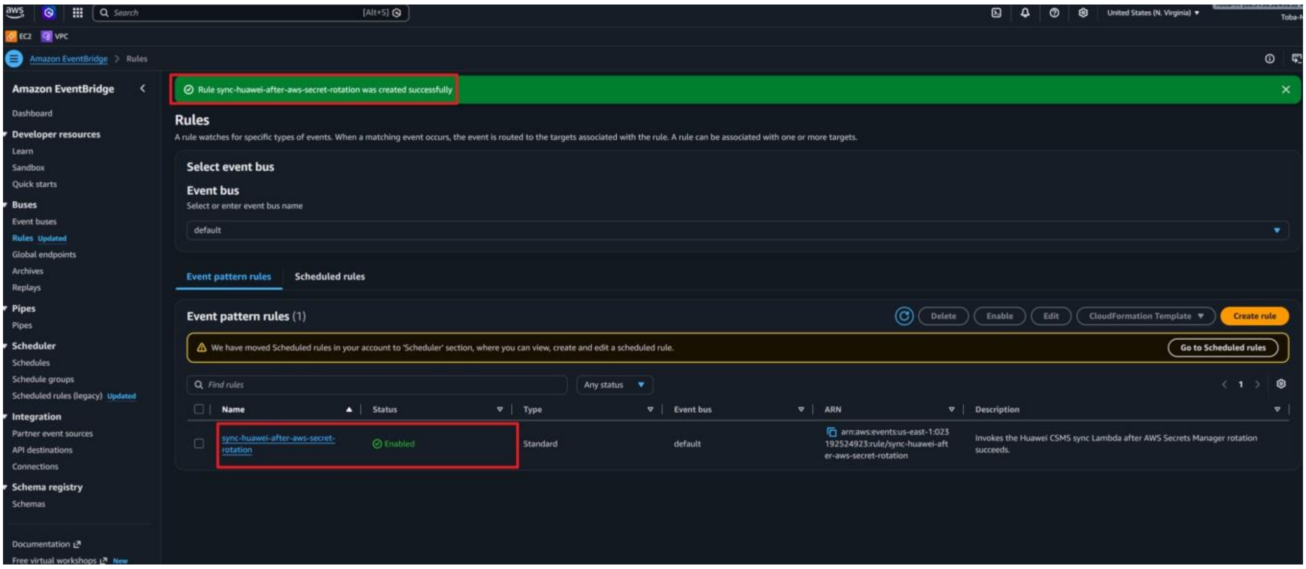
Set the target to **Lambda function**.

Choose the function `sync-aws-secret-to-huawei-csms`.

Click **Next**, skip tags, then click **Create rule**.







Step 4 — Add explicit invoke permission for EventBridge

Run this in CloudShell:

```
RULE_ARN=$(aws events describe-rule \
  --name sync-huawei-after-aws-secret-rotation \
  --query Arn --output text --region us-east-1)
aws lambda add-permission \
  --function-name sync-aws-secret-to-huawei-csms \
  --statement-id AllowEventBridgeInvokeSync \
  --action lambda:InvokeFunction \
  --principal events.amazonaws.com \
  --source-arn "$RULE_ARN" \
  --region us-east-1
```

STAGE 9
CREATE THE CLOUDTRAIL TRAIL

> **NOTE — Why a CloudTrail trail is required**
> CloudTrail Event History shows events in the console, but EventBridge only receives CloudTrail-based events when an active trail is streaming them.
> Without this trail, EventBridge will never see the UpdateSecretVersionStage event and will never invoke the sync Lambda.
> This was the missing piece that caused the first EventBridge test to fail.

Step 1 — Create the trail

Go to **CloudTrail** → **Trails** → **Create trail**.

Set the trail name to `secret-rotation-eventbridge-trail`.

Choose **Create new S3 bucket**.

Enable **Multi-region trail**.

Under Management events, enable **Write events**.

Leave Insights disabled.

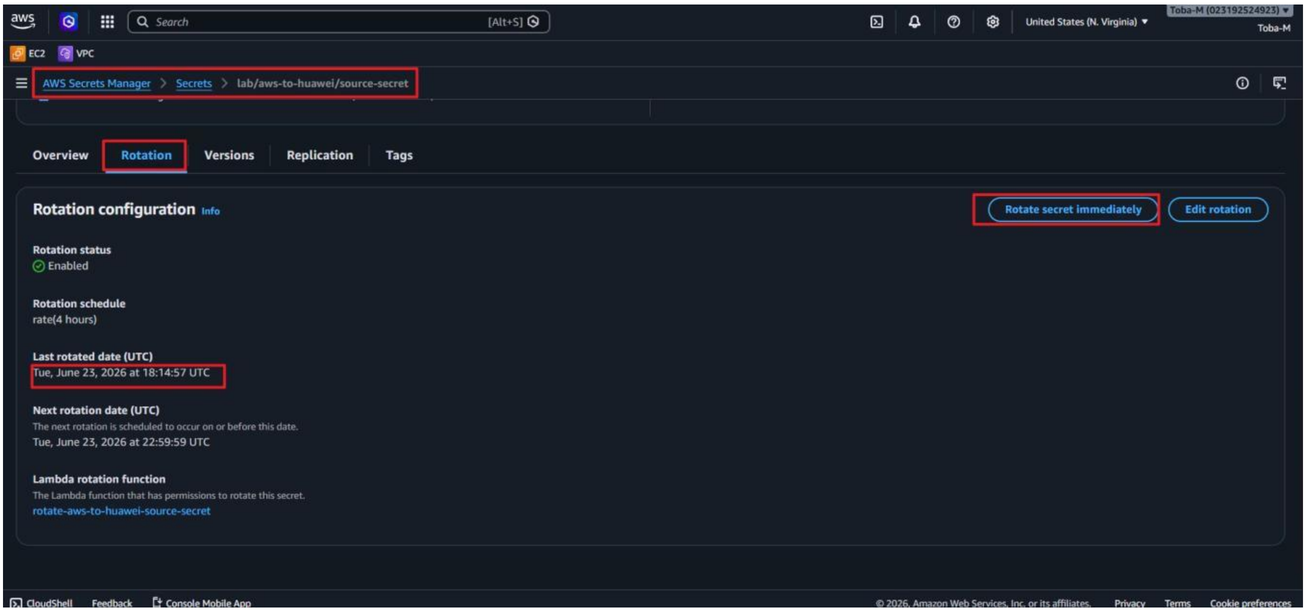
Click **Create trail**.

STAGE 10
RUN THE END-TO-END TEST

Step 1 — Trigger an immediate rotation

Go to **AWS Secrets Manager** → `lab/aws-to-huawei/source-secret` → **Rotation**.

Click **Rotate secret immediately**.



The time in **AWS is like an hour behind.**

Step 2 — Watch the CloudWatch logs

Wait 1 to 3 minutes, then run this in CloudShell:

```
aws logs tail /aws/lambda/sync-aws-secret-to-huawei-csms \
  --since 15m --region us-east-1
```

The successful output looks like this:

```
2026-06-23T18:15:00.893Z   Starting AWS to Huawei CSMS secret sync.
2026-06-23T18:15:02.487Z   Secret synced to Huawei CSMS.
                           HuaweiSecretName=aws-huawei_replica
                           HuaweiVersionId=v4
```

SUCCESS CHECK

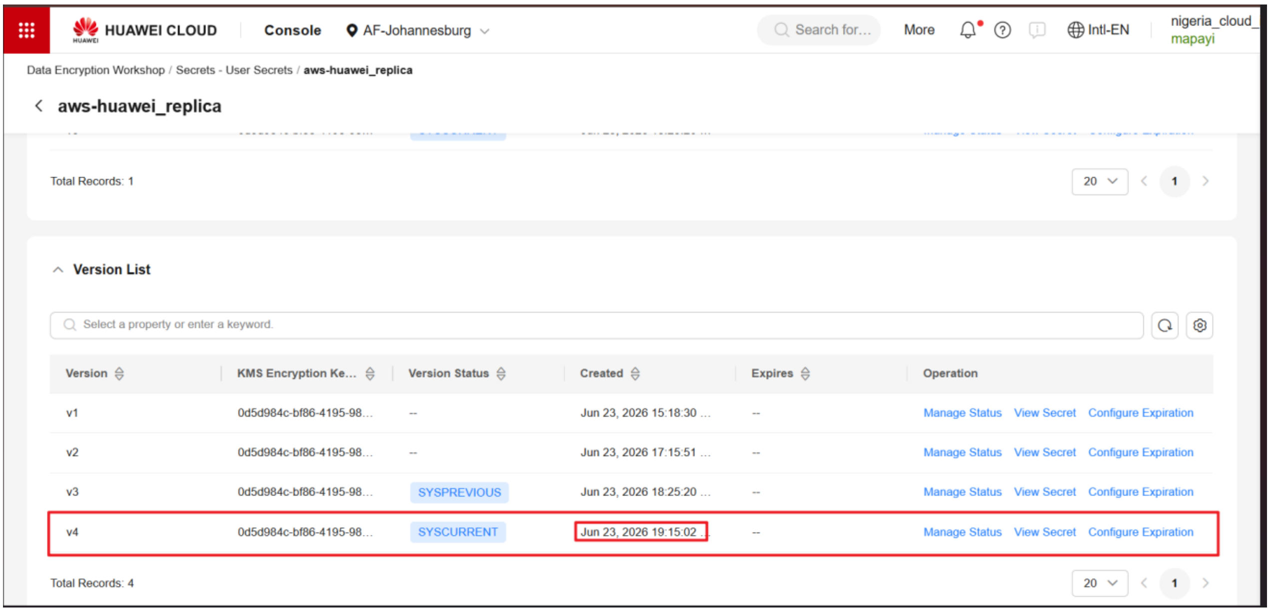
The log entry "Secret synced to Huawei CSMS" confirms the full pipeline is working.
EventBridge received the CloudTrail event, invoked the sync Lambda, and the Lambda created a new Huawei CSMS version.

Step 3 — Confirm the new version in Huawei CSMS

Open Huawei Cloud console → **DEW** → **CSMS** → `aws-huawei_replica`.

Confirm a new version exists with the stage `SYSCURRENT`.

The previous version should show `SYSPREVIOUS`.


HUAWEI CLOUD
Console
AF-Johannesburg
Search for...
More
Intl-EN
nigeria_cloud_
mapayi
Data Encryption Workshop / Secrets - User Secrets / aws-huawei_replica
aws-huawei_replica
Total Records: 1
Version List
Select a property or enter a keyword.
Version
KMS Encryption Ke...
Version Status
Created
Expires
Operation
v1
0d5d984c-bf86-4195-98...
--
Jun 23, 2026 15:18:30 ...
Manage Status
View Secret
Configure Expiration
v2
Jun 23, 2026 17:15:51 ...
v3
SYSPREVIOUS
Jun 23, 2026 18:25:20 ...
v4
SYSCURRENT
Jun 23, 2026 19:15:02
Total Records: 4

What Finally Made It Work

> **NOTE — Key learning from this lab**
> CloudTrail Event History shows Secrets Manager events in the AWS console, but EventBridge routing requires an active, streaming CloudTrail trail.
> Without the trail, EventBridge never saw the UpdateSecretVersionStage event, and the sync Lambda was never invoked.
> Creating the trail was the missing piece.

The complete working flow is:

```
AWS Secrets Manager rotation triggered
    ↓
UpdateSecretVersionStage API event fires
    ↓
CloudTrail trail streams the event to EventBridge
    ↓
EventBridge rule matches the event pattern
    ↓
sync-aws-secret-to-huawei-csms Lambda invoked
    ↓
Huawei CSMS receives new secret version — SYSCURRENT
updated
```

What to Do If Something Goes Wrong

Problem 1 — Secrets Manager cannot invoke the rotation Lambda

Error message: *"Secrets Manager can't invoke the specified Lambda function."*

Cause: The rotation Lambda does not have a resource-based policy allowing Secrets Manager to invoke it.

Fix: Run the `aws lambda add-permission` CloudShell command in Stage 3, Step 3 of this guide.

Problem 2 — Sync Lambda fails with HostUnreachableException

Error message: *"Failed to resolve csms.af-south-1.myhuaweicloud.com"*

Cause: The HUAWEI_CSMS_ENDPOINT environment variable is set to the wrong hostname.

Fix: Change the variable to `https://kms.af-south-1.myhuaweicloud.com` and redeploy.

Problem 3 — EventBridge does not invoke the sync Lambda after rotation

Symptom: The manual sync test works, but no new Huawei CSMS version appears after rotation.

Cause: No active CloudTrail trail exists, so EventBridge never receives the rotation event.

Fix: Create the CloudTrail trail as described in Stage 9. Then rotate the secret again.

Problem 4 — put-events returns NotAuthorizedForSourceException

Error message: *"Not authorized for the source."*

Cause: AWS prevents users from manually spoofing events with the reserved source aws.secretsmanager.

Fix: This is expected behaviour. Trigger a real rotation using **Rotate secret immediately** instead.

Problem 5 — Sync Lambda times out

Symptom: The Lambda invocation fails with a timeout error.

Cause: The default Lambda timeout of 3 seconds is too short for cross-cloud API calls.

Fix: Go to Configuration → General configuration → Edit and set the timeout to 30 seconds.

Quick Recap

Here is what you accomplished in this lab:

- Created a custom key-value secret in AWS Secrets Manager.
- Built a rotation Lambda in Python that generates a new password every 4 hours.
- Created an IAM inline policy granting the rotation Lambda access to the source secret.
- Granted Secrets Manager permission to invoke the rotation Lambda using aws lambda add-permission.
- Enabled automatic rotation on the source secret.
- Created a target secret in Huawei Cloud DEW CSMS in the AF-Johannesburg region.
- Built a sync Lambda that reads the AWS AWSCURRENT value and creates a new Huawei CSMS version.
- Packaged the Huawei Python SDK as a Lambda Layer and attached it to the sync Lambda.
- Added environment variables including the correct CSMS endpoint: kms.af-south-1.myhuaweicloud.com.
- Created an EventBridge rule that fires on UpdateSecretVersionStage events from CloudTrail.
- Created a CloudTrail trail — the critical missing piece that enables EventBridge routing.
- Ran a successful end-to-end test: rotation triggered a new Huawei CSMS version automatically.

Most important things to remember

- The **correct Huawei CSMS endpoint** for AF-Johannesburg is `https://kms.af-south-1.myhuaweicloud.com` — not csms.
- EventBridge can only route CloudTrail events when an **active CloudTrail trail** exists. Event History alone is not enough.
- The **resource-based Lambda permission** (aws lambda add-permission) is required separately for both Secrets Manager and EventBridge.
- Never store Huawei AK/SK in Lambda environment variables in production. Use AWS Secrets Manager or a secure identity federation method instead.

Cleanup

Run these commands in CloudShell to remove all lab resources when you are done.

Delete the AWS source secret

```
# Safe delete — 7-day recovery window
aws secretsmanager delete-secret \
  --secret-id lab/aws-to-huawei/source-secret \
  --recovery-window-in-days 7 \
  --region us-east-1
```

Delete the Lambda functions and layer

```
aws lambda delete-function --function-name rotate-aws-to-huawei-source-secret --region
us-east-1
aws lambda delete-function --function-name sync-aws-secret-to-huawei-csms --region us-
east-1
```

Remove the EventBridge rule and target

```
TARGET_ID=$(aws events list-targets-by-rule \
  --rule sync-huawei-after-aws-secret-rotation \
  --region us-east-1 --query "Targets[0].Id" --output text)
aws events remove-targets \
  --rule sync-huawei-after-aws-secret-rotation \
  --ids "$TARGET_ID" --region us-east-1
aws events delete-rule \
  --name sync-huawei-after-aws-secret-rotation --region us-east-1
```

Delete the CloudTrail trail

```
aws cloudtrail delete-trail --name secret-rotation-eventbridge-trail --region us-east-
1
```

NOTE
Empty and delete the CloudTrail S3 bucket manually from the S3 console after deleting the trail.

Delete the Huawei CSMS secret

Go to **Huawei Console** → **Data Encryption Workshop** → **Cloud Secret Management Service** → `aws-huawei_replica` → **Delete secret**.